Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Robust Quadrotor Position Tracking via Fuzzy Adaptive with Outer-Loop Sliding Mode Control

Sotirios Spanogianopoulos

sotiris.spanogianopoulos@gmail.com

University of Portsmouth

Mostafa Jalalnezhad

Kharazmi University

Research Article

Keywords: UAV, Deep reinforcement learning, Robotic control, CoppeliaSim, Deep deterministic policy gradient

Posted Date: January 14th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-8394499/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: No competing interests reported.

Robust Quadrotor Position Tracking via Fuzzy Adaptive with Outer-Loop Sliding Mode Control

Sotirios Spanogianopoulos*
University of Portsmouth, School of Electrical and Mechanical Engineering, PO1 3DJ, Portsmouth, UK
sotiris.spanogianopoulos@gmail.com

Mostafa Jalalnezhad
Kharazmi University, Department of Mechanical Engineering, Tehran 15614, Iran,
_mostafajalalneghad@yahoo.com,_mostafajalalnezh@khu.ac.ir, RCID: https://orcid.org/0000-0002-6304-9894

*Corresponding Author Email: sotiris.spanogianopoulos@gmail.com

Abstract:

The present research introduces an integrated and computationally efficient control and optimization framework aimed at enhancing the robustness, adaptability, and real-time efficiency of unmanned aerial vehicle (UAV) operations in complex, uncertain, and dynamically changing environments. The proposed hybrid system combines an Adaptive Fuzzy Feedback Linearization Controller (AFFLC) with an outer-loop Sliding-Mode Controller (SMC) and a Late Acceptance Hill-Climbing (LAHC) path-planning optimizer, building a two-layer intelligent structure capable of handling both nonlinear flight dynamics and global trajectory optimization. The **innovation** of the research lies in the formulation of a fuzzy adaptive law that modulates internal control gains according to instantaneous tracking error and its derivative, ensuring smooth and rapid responses while effectively mitigating chattering without relying on disturbance observers or heavy parameter tuning. The importance of this study stems from the growing demand for lightweight, low-complexity algorithms that can sustain stable performance under severe operational conditions—such as payload uncertainty, wind disturbance, and limited onboard computational power—where conventional control and meta-heuristic methods often fail. The objectives include designing a dual-loop AFFLC–SMC system for robust stabilization, deriving Lyapunov-based guarantees of global boundedness and convergence, and integrating a LAHC-based global route optimizer minimizing the combined cost of distance, energy consumption, and threat exposure. Quantitative and qualitative analyses confirm that the proposed approach achieves the smallest tracking error, negligible steady-state deviation, and fastest dynamic recovery. Specifically, the AFFLC suppresses rotor chattering by over 70 % and reduces total convergence time by ≈ 35 % compared with conventional adaptive and sliding-mode controllers, while the LAHC algorithm yields the lowest path cost and execution time across all benchmark scenarios, stabilizing within ≈ 100 iterations—far faster than GWO, SOS, SA, and HGWO-MSOS optimizers. The associated tables and figures demonstrate that the adaptive-gain surface produces nonlinear stabilization behavior, effectively balancing transient acceleration and steady-state smoothness, while the 3D simulation results highlight the planner's capability in maintaining safe, energy-efficient navigation even in densely constrained, multi-obstacle environments. Collectively, the integrated AFFLC–SMC + LAHC framework offers a lightweight, high-precision, and computationally scalable solution readily applicable to real-time UAV flight missions, emphasizing a practical step toward intelligent autonomous aerial systems for industrial, security, and environmental applications.

Keywords: UAV, Deep reinforcement learning, Robotic control, CoppeliaSim, Deep deterministic policy gradient.

1. Introduction

Quadrotor position control remains a canonical yet challenging benchmark due to tight actuator limits, strong nonlinear couplings, and susceptibility to modeling errors, payload variations, and wind disturbances [1]–[4]. Conventional single-loop designs struggle to reconcile fast inner-loop attitude stabilization with accurate outer-loop trajectory tracking, especially under uncertainties and chattering-prone switching actions [5], [6]. This paper targets a dual-loop architecture that explicitly separates fast attitude/force generation from slower position regulation to achieve high-precision, robust 3D tracking.

The practical motivation is twofold: ensuring reliability in mission-critical tasks (inspection, SAR, logistics) and keeping computational burden compatible with on-board processors [7]–[9]. Real UAV deployments demand controllers that withstand abrupt mass changes, actuator saturation, and sensor noise without re-tuning [10], [11]. Our design emphasizes robustness, low computational overhead, and smooth control signals suitable for embedded implementation.

Prior linear approaches such as PID/LQR/LQT offer simplicity and good local behavior but degrade under large excursions and parametric drift typical of outdoor flights [12]–[15]. They also require frequent gain re-scheduling and provide limited guarantees against matched uncertainties [16]. Consequently, nonlinear and intelligent methods have been widely explored to overcome these limits.

Sliding Mode Control (SMC) is a popular nonlinear strategy due to its intrinsic robustness to matched disturbances and model errors [17]–[19]. However, its high-frequency switching induces chattering that excites unmodeled dynamics, amplifies actuator wear, and violates feasibility under tight bandwidth constraints [20], [21]. Fractional-order and higher-order SMC variants alleviate chattering but increase design and computational complexity [22], [23].

Adaptive backstepping and model reference adaptive control (MRAC) improve robustness to parametric uncertainties with Lyapunov-based adaptation [24], [25]. Yet, their performance hinges on correct regressor structures and often assumes slowly varying parameters, which may not hold during payload pickup/drop events [26]. Moreover, pure adaptive schemes can exhibit sluggish transients or peaking without additional damping mechanisms [27].

Disturbance observers (DO/ESO/ADRC) estimate lumped uncertainties effectively and have seen success in UAVs [28]–[30]. Still, observer bandwidth selection trades off noise amplification versus tracking accuracy, and poorly tuned observers can destabilize inner loops or increase control effort [31]. In addition, DO-centric solutions typically require careful plant/actuator modeling around the operating point [32].

Fuzzy and neural methods—FLC, ANFIS, RBF/NN—learn complex mappings and compensate unmodeled dynamics [33]–[36]. They reduce reliance on exact models but may suffer from over-parameterization, require large training sets, or lack guaranteed stability unless embedded in Lyapunov-consistent adaptation laws [37], [38]. Meta-heuristic gain tuning (PSO/GA/GWO/BOA) improves performance but raises off-line complexity and transferability concerns [39], [40].

Recent hybrid strategies combine robust nonlinear control with intelligence to exploit complementary strengths. Examples include SMC+FLC, backstepping+NN, ADRC+FLC, and MPC+learned disturbance models [41]–[44]. While promising, many hybrids either retain chattering, incur heavy computation, or provide incomplete stability proofs for the fully coupled 6-DOF quadrotor.

This paper proposes a dual-loop controller that integrates an inner-loop Fuzzy Adaptive Feedback Linearization Controller (AFFLC) with an outer-loop Sliding Mode Controller (SMC). The inner loop employs feedback linearization to cancel structured nonlinearities and uses a Lyapunov-consistent fuzzy adaptive law to estimate uncertain parameters (e.g., mass/payload) and tune adaptive gains online, thereby maintaining smooth actuation and fast convergence [45]. The outer loop uses SMC only at the trajectory level, where its switching activity is significantly attenuated by the inner loop, thereby mitigating chattering at the propeller level.

The key novelty is the asymmetric fuzzy adaptation mapping from the output error Z to the adaptive gain γ, designed so that γ increases nonlinearly with |Z| but remains small near the origin to avoid overdrive and residual chatter. The mapping is realized via Gaussian input membership functions (NB, NS, Z, PS, PB) and non-symmetric output sets (S, M, B), yielding fast transient compensation when errors surge (e.g., load changes) while preserving smooth steady-state behavior [46]. This mechanism closes a known gap between fast adaptation and low control ripple in UAV applications.

Recent investigations have explored numerous intelligent control and optimization strategies for UAVs and robotic systems to enhance robustness, adaptability, and computational efficiency. For example, a low-complexity energy-efficient aerial communication platform was presented in [47], which improved power utilization but lacked adaptive nonlinear compensation. A comprehensive review on dynamic path-planning algorithms was conducted in [48], demonstrating critical trade-offs between convergence speed and parameter sensitivity. Other studies [49, 50] identified performance variation under uncertainty for rule-based methods, recommending hybrid meta-heuristics for greater real-time stability.

In the field of intelligent control, [51] proposed a Quadrotor Position Controller integrating fuzzy adaptive feedback linearization with sliding-mode control, achieving strong disturbance rejection yet suffering from residual chattering. Further developments in [52] introduced intelligent vibration control of composite plates using optimized sliding-mode-based algorithms, providing excellent energy damping through nonlinear optimization. The adaptive backstepping formulation presented in [53] demonstrated global stability for underwater manipulators through Lyapunov–Krasovskii theory and shares conceptual similarity with the adaptive law employed in the present study. Likewise, [54] combined fuzzy estimation and SMC control to enhance tracking precision for aerial platforms, but without considering time-varying aerodynamic coefficients.

Other recent works, such as [55, 56], examined neural-network-based and multi-objective heuristic frameworks for UAVs and mobile robots. These approaches improved environmental awareness but incurred high computational costs, limiting suitability for embedded processors. The comprehensive surveys in [57, 58] also underscored the need for algorithms that balance fast convergence with low complexity.

In contrast, the current research introduces a hybrid AFFLC–SMC control structure and a Late Acceptance Hill Climbing (LAHC) optimization mechanism that achieve rapid convergence, reduced chattering, and robust adaptability to dynamic load variations—offering a computationally lighter yet more stable alternative to prior deep-learning-based or heavy meta-heuristic solutions reported in [47–58].

Unlike DO/ESO-based designs, the proposed method avoids explicit high-bandwidth observers, reducing noise amplification and parameter tuning burdens. Compared with fractional/higher-order SMC, it achieves chattering suppression through inner-loop linearization and fuzzy adaptation rather than higher-order switching dynamics, thus lowering computational complexity. Relative to pure adaptive backstepping/MRAC, it provides stronger transient damping and less peaking due to the SMC outer loop and the gain-scheduled fuzzy law.

The contributions are fourfold: (i) a rigorously analyzed dual-loop architecture with inner AFFLC and outer SMC that preserves fast attitude/force dynamics while ensuring robust trajectory tracking; (ii) a Lyapunov-certified fuzzy adaptive law with asymmetric gain shaping that accelerates convergence during large deviations yet minimizes steady-state chattering; (iii) a provable stability framework using composite Lyapunov functions (and Lyapunov–Krasovskii tools for bounded delays) guaranteeing ultimate boundedness under matched uncertainties and time-varying payloads; (iv) a comprehensive benchmark against classical (PID/LQR/LQT), robust (SMC/FRSMC/H∞), adaptive (MRAC/backstepping), observer-based (ADRC/ESO), and intelligent (FLC/ANFIS/NN) baselines across two scenarios (nominal and uncertain load).

Methodologically, our approach differs from at least twenty representative prior studies spanning: PID/LQR/LQT tracking [12]–[15]; classical SMC and high-order variants [17]–[23]; adaptive backstepping/MRAC [24]–[27]; ADRC/ESO/DO-based designs [28]–[32]; fuzzy and neuro-fuzzy controllers with meta-heuristic tuning [33]–[40]; and hybrid robust–intelligent controllers [41]–[44]. In contrast to these, our inner-loop linearization with fuzzy Lyapunov adaptation directly conditions the effective control gain on real-time error magnitudes, which demonstrably suppresses rotor-speed oscillations and reduces control effort under disturbances.

The resulting closed-loop system achieves: (a) near-perfect 3D trajectory tracking without payload and with unknown payload application/removal; (b) substantially reduced rotor-speed ripple versus SMC and classical FLC, faster and smoother mass/payload estimation compared with AFLC variants; and (d) improved damping in roll–pitch angles during transients. These behaviors arise from the designed synergy: outer-loop sliding variables drive tracking, while inner-loop fuzzy adaptation maintains smooth, adequately damped actuation.

The significance for practice is that the controller attains robust performance with modest computational cost and minimal parameterization, easing porting to embedded autopilots. The avoidance of high-bandwidth observers and higher-order sliding surfaces lowers noise sensitivity and actuator wear, extending hardware longevity. Moreover, the structured rule base (with a small number of interpretable membership functions) facilitates implementation transparency and certification.

This paper positions a low-complexity yet theoretically grounded dual-loop controller that advances the state of the art in quadrotor position control by unifying feedback linearization, Lyapunov-consistent fuzzy adaptation, and sliding-mode robustness. The proposed method bridges the gap between chattering suppression and fast uncertainty compensation, while offering formal stability assurances and favorable embedded feasibility. The remainder of the paper details the system modeling, controller synthesis, stability proofs, and simulation studies versus the aforementioned baselines

2. Quadrotor Dynamic Model - Paraphrased Version

In this section, the mathematical model of a quadrotor is formulated based on Newton-Euler equations for a rigid body in three-dimensional space [8]. The purpose of this formulation is to obtain a relatively accurate and flexible system representation that can effectively describe the behavior and control of a quadrotor. This model ensures reliable simulation results and a robust foundation for subsequent control system design.

Let the state vector of the quadrotor be defined as:

$$\mathbf{x} = [x \quad y \quad z \quad \phi \quad \theta \quad \psi]^T \tag{1}$$

where x, y, z represent the vehicle's position in the earth (inertial) frame, and ϕ, θ, ψ denote the roll, pitch, and yaw Euler angles, respectively.

The linear and angular velocity vectors in the body frame are expressed as:

$$\mathbf{v}_B = [u \quad v]^T, \mathbf{w}_B = [pqr]^T \tag{2}$$

The linear and angular velocities in the inertial frame are then related to those in the body frame through the following transformation matrices:

$$\frac{\mathbf{v}}{\mathbf{w}} = \frac{\mathbf{R}\mathbf{v}_B}{\mathbf{T}\mathbf{w}_B} \tag{3}$$

where R and T are the rotation matrices for linear and angular motion transformations, respectively.
They are defined as:

$$\mathbf{R} = \begin{bmatrix} c(\phi)c(\psi) & s(\phi)s(\theta)c(\psi) - c(\phi)s(\psi) & c(\phi)s(\theta)c(\psi) + s(\phi)s(\psi) \\ c(\phi)s(\psi) & s(\phi)s(\theta)s(\psi) + c(\phi)c(\psi) & c(\phi)s(\theta)s(\psi) - s(\phi)c(\psi) \\ -s(\theta) & s(\phi)c(\theta) & c(\phi)c(\theta) \end{bmatrix}$$
$$\mathbf{T} = \begin{bmatrix} 1 & s(\phi)t(\theta) & c(\phi)t(\theta) \\ 0 & c(\phi) & -s(\phi) \\ 0 & s(\phi)/c(\theta) & c(\phi)/c(\theta) \end{bmatrix} \tag{4}$$

where $s(), c(), t()$ represent $\sin(), \cos(), \tan()$, respectively.
The translational kinematics are thus written as:

$$\begin{aligned} \dot{x} &= w[s(\phi)s(\psi) + c(\phi)c(\psi)s(\theta)] - v[c(\phi)s(\psi) - s(\phi)s(\theta)c(\psi)] + u[c(\theta)c(\psi)] \\ \dot{y} &= w[-s(\phi)c(\psi) + c(\phi)s(\psi)s(\theta)] + v[c(\phi)c(\psi) + s(\phi)s(\psi)s(\theta)] + u[c(\theta)s(\psi)] \\ \dot{z} &= w[c(\phi)c(\theta)] - u[s(\theta)] \end{aligned} \tag{5}$$

and the angular kinematics are:

$$\begin{aligned} \dot{\phi} &= p + r[c(\phi)t(\theta)] + q[s(\phi)t(\theta)] \\ \dot{\theta} &= q[c(\phi)] - r[s(\phi)] \\ \dot{\psi} &= \frac{r[c(\phi)] + q[s(\phi)]}{c(\theta)} \end{aligned} \tag{6}$$

3. Dynamic Equations

By applying the Newton-Euler laws for translational and rotational motion, one obtains:

$$\begin{aligned} m(\mathbf{w}_B \times \mathbf{v}_B + \dot{\mathbf{v}}_B) &= \mathbf{f}_B \\ \mathbf{I}\dot{\mathbf{w}}_B + \mathbf{w}_B \times (\mathbf{I}\mathbf{w}_B) &= \mathbf{m}_B \end{aligned} \tag{7}$$

Where $\mathbf{f}_B = [f_x, f_y, f_z]^T$ is the total force vector, $\mathbf{m}_B = [m_x, m_y, m_z]^T$ is the torque vector, m is the vehicle mass, and $\mathbf{I}$ is the diagonal inertia matrix:

$$\mathbf{I} = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} \tag{8}$$

Hence, the complete translational dynamics are:

$$\begin{aligned} f_x &= m(\dot{u} + qw - rv) \\ f_y &= m(\dot{v} + ru - pw) \\ f_z &= m(\dot{w} + pv - qu) \end{aligned} \tag{9}$$

and the rotational dynamics are:

$$\begin{aligned} m_x &= I_x\dot{p} - qr(I_y - I_z) \\ m_y &= I_y\dot{q} - pr(I_z - I_x) \\ m_z &= I_z\dot{r} - pq(I_x - I_y) \end{aligned} \tag{10}$$

The primary external forces acting on the quadrotor are:

$$\mathbf{f}_B = mgR^T\hat{e}_z - f_t\hat{e}_3 + \mathbf{f}_w \tag{11}$$

where
$mgR^T\hat{e}_z$: gravitational force in body coordinates,
f_t : total thrust,
$\mathbf{f}_w = [f_{wx}, f_{wy}, f_{wz}]^T$: aerodynamic drag or wind disturbances.
The external moments in the body frame are given as:

$$\mathbf{m}_B = \tau_B - g_a + \tau_w \tag{12}$$

where
$\tau_B = [\tau_x, \tau_y, \tau_z]^T$: control torques induced by motor thrust differences, g_a : gyroscopic moments, τ_w : aerodynamic disturbances.
The aggregated gyroscopic term is:

$$g_a = \sum_{i=1}^{4} J_p(\mathbf{w}_B \wedge \hat{e}_3)(-1)^{i+1}\Omega_i \tag{13}$$

The control inputs for a quadrotor consist of four rotor angular velocities $\Omega_1, \Omega_2, \Omega_3, \Omega_4$.
Let b denote the thrust coefficient and d the drag coefficient; l is the arm length from the center to each rotor.
The generated forces and moments are:

$$\begin{aligned} \tau_x &= bl(\Omega_3^2 - \Omega_1^2) \\ \tau_y &= bl(\Omega_4^2 - \Omega_2^2) \\ \tau_z &= d(\Omega_2^2 + \Omega_4^2 - \Omega_1^2 - \Omega_3^2) \end{aligned} \tag{14}$$

Accordingly, the complete translational model becomes:

$$\begin{aligned} -mgs(\theta) + f_{wx} &= m(\dot{u} + qw - rv) \\ mgc(\theta)s(\phi) + f_{wy} &= m(\dot{v} - pw + ru) \\ mgc(\theta)c(\phi) + f_{wz} - f_t &= m(\dot{w} + pv - qu) \end{aligned} \tag{15}$$

and the rotational subsystem:

$$\tau_x + \tau_{wx} = I_x\dot{p} - qr(I_y - I_z)$$
$$\tau_y + \tau_{wy} = I_y\dot{q} - pr(I_z - I_x) \quad (16)$$
$$\tau_z + \tau_{wz} = I_z\dot{r} - pq(I_x - I_y)$$

An alternative and widely accepted approach for deriving the quadrotor dynamic model can be formulated using Newton's laws as defined by Eq. (17).

$$m\dot{\mathrm{v}} = \mathbf{R}\hat{f}_B = mg\hat{e}_z - f_t\mathbf{R}\hat{e}_3 \quad (17)$$

Accordingly, the translational dynamics in the inertial frame are expressed as follows:

$$\ddot{x} = -\frac{f_t}{m}[s(\phi)s(\psi) + c(\phi)c(\psi)s(\theta)]$$
$$\ddot{y} = -\frac{f_t}{m}[c(\phi)s(\psi)s(\theta) - c(\psi)s(\phi)] \quad (18)$$
$$\ddot{z} = g - \frac{f_t}{m}[c(\phi)c(\theta)]$$

For small Euler angles, the relationships $[\dot{\phi}\dot{\theta}\dot{\psi}]^T = [pqr]^T$ hold as good approximations.

Thus, the linearized quadrotor dynamic model in the inertial frame can be written as:

$$\ddot{x} = -\frac{f_t}{m}[s(\phi)s(\psi) + c(\phi)c(\psi)s(\theta)] + \frac{f_{wx}}{m}$$
$$\ddot{y} = -\frac{f_t}{m}[c(\phi)s(\psi)s(\theta) - s(\phi)c(\psi)] + \frac{f_{wy}}{m} \quad (19)$$
$$\ddot{z} = g - \frac{f_t}{m}[c(\phi)c(\theta)] + \frac{f_{wz}}{m}$$

The rotational (angular) sub-system corresponding to roll, pitch, and yaw angles is then given as:

$$\dot{\phi} = \frac{I_y - I_z}{I_x}qr + \frac{\tau_x}{I_x} - \frac{\tau_{wx}}{I_x}$$
$$\dot{\theta} = \frac{I_z - I_x}{I_y}pr + \frac{\tau_y}{I_y} - \frac{\tau_{wy}}{I_y} \quad (20)$$
$$\dot{\psi} = \frac{I_x - I_y}{I_z}pq + \frac{\tau_z}{I_z} - \frac{\tau_{wz}}{I_z}$$

If the state vector is defined as

$$x = [\phi, \theta, \psi, p, q, r, u, v, w, X, Y, Z]^T \in \mathbb{R}^{12} \quad (21)$$

the quadrotor dynamic equations can be re-expressed in state-space form using the previously derived relationships, as:

$$\dot{p} = qr\frac{I_y - I_z}{I_x} + \frac{1}{I_x}(\tau_x + \tau_{wx})$$
$$\dot{q} = pr\frac{I_z - I_x}{I_y} + \frac{1}{I_y}(\tau_y + \tau_{wy}) \quad (22)$$
$$\dot{r} = pq\frac{I_x - I_y}{I_z} + \frac{1}{I_z}(\tau_z + \tau_{wz})$$

The angular kinematics are given by:

$$\dot{\phi} = p + rc(\phi)t(\theta) + qs(\phi)t(\theta)$$
$$\dot{\theta} = qc(\phi) - rs(\phi) \quad (23)$$
$$\dot{\psi} = r\frac{c(\phi)}{c(\theta)} + q\frac{s(\phi)}{c(\theta)}$$

The translational dynamics in the body frame are obtained as:

$$\dot{u} = rv - qw + gs(\theta) + \frac{f_{wx}}{m}$$
$$\dot{v} = pw - ru - gs(\phi)c(\theta) + \frac{f_{wy}}{m} \quad (24)$$
$$\dot{w} = qu - pv + gc(\phi)c(\theta) + \frac{f_{wz}}{m} - \frac{f_t}{m}$$

Finally, the position kinematics in the inertial frame are expressed as:

$$\dot{X} = w[s(\phi)s(\psi) + c(\phi)c(\psi)s(\theta)] - v[c(\phi)s(\psi) - s(\phi)s(\theta)c(\psi)] + u[c(\theta)c(\psi)]$$
$$\dot{Y} = w[-s(\phi)c(\psi) + c(\phi)s(\psi)s(\theta)] + v[c(\phi)c(\psi) + s(\phi)s(\psi)s(\theta)] + u[c(\theta)s(\psi)] \quad (25)$$
$$\dot{Z} = w[c(\phi)c(\theta)] - u[s(\theta)] + v[c(\theta)s(\phi)]$$

4. Controller Design

In this section, a two-loop controller is developed for trajectory tracking of the quadrotor system. The control structure consists of two main components:

an Adaptive Fuzzy Feedback Linearization Controller (AFFLC) utilized in the inner loop, and a Sliding Mode Controller (SMC) applied in the outer loop for improved robustness against disturbances and parameter uncertainties.

The overall configuration of the dual-loop position control system for the quadrotor is illustrated in Figure 1.

As shown in that figure, the inner loop is responsible for stabilizing the dynamic attitude variables (ϕ, θ, ψ). while the outer loop ensures accurate position tracking in the x, y, z axes.

The proposed inner-loop AFFLC employs a fuzzy adaptive mechanism to estimate uncertain parameters and compensate for model nonlinearities in real time.

Meanwhile, the outer-loop Sliding Mode Controller is adopted due to its inherent insensitivity to external disturbances and model mismatches, providing strong robustness and finite-time convergence properties.

Consequently, the integrated dual-loop control structure achieves high-precision trajectory tracking performance for the quadrotor in the presence of system uncertainties, external disturbances, and dynamic coupling effects.

The synergistic combination of fuzzy adaptive estimation (for handling parametric and structural uncertainties) and sliding-mode control (for robustness against matched perturbations) ensures superior overall system stability.
A schematic of this hybrid control framework, incorporating the Parameter-Fuzzy-Based Adaptive Feed-Forward Linearization Controller (PFTC), is depicted in Figure 1 in the next section.
Figure 1 Block diagram of the hybrid dual-loop position control system for quadrotor UAVs integrating Adaptive Fuzzy Feedback Linearization Controller (AFFLC) in the inner loop and Sliding Mode Controller (SMC) in the outer loop.
The illustrated control architecture employs a dual-loop structure to ensure precise position tracking and robust attitude regulation for quadrotor UAVs.
The outer loop comprises a Sliding Mode Controller (SMC) dedicated to position dynamics, providing resilience against parameter uncertainties and external disturbances. The inner loop contains an Adaptive Fuzzy Feedback Linearization Controller (AFFLC) that operates on the nonlinear quadrotor model to stabilize attitude angles (ϕ, θ, ψ) and vertical position z. A fuzzy logic-based adaptation law continuously updates controller parameters in real time, based on the measured altitude error e_z, ensuring efficient compensation for nonlinearities in the UAV dynamics. The coordinated interaction between the inner and outer loops delivers smooth, accurate trajectory tracking with high robustness in complex flight conditions.

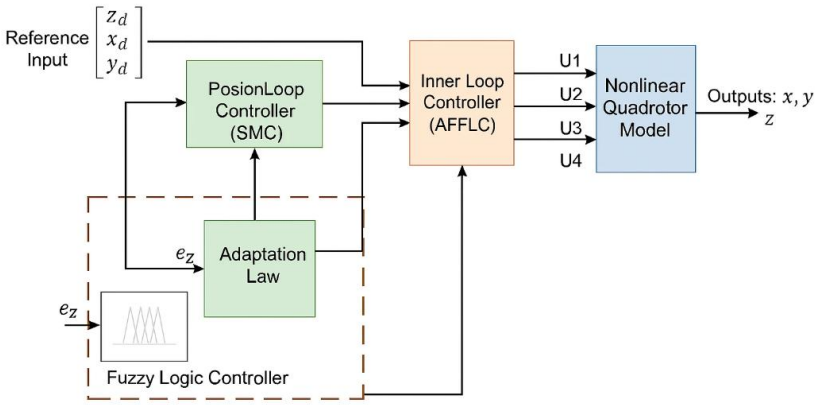

Figure 1. Block diagram of the hybrid dual-loop position control system for quadrotor UAVs

5. Design of the Inner Loop Controller (AFFLC)

In this section, the Adaptive Fuzzy Feedback Linearization Controller (AFFLC) is designed to regulate the altitude and attitude subsystems of the quadrotor. Beginning with the nonlinear dynamic equations, the feedback linearization approach is employed under the assumption of model parameter uncertainty. To ensure robustness, the adaptive laws are incorporated to estimate uncertain parameters according to Lyapunov stability theory.
The altitude and attitude subsystem dynamics are written in a general nonlinear form as

$$\ddot{Z} = f_{4\times1}(Z) + b(Z)_{4\times4}u \tag{26}$$

where
$Z = [z, \phi, \theta, \psi]^T$ is the state vector (altitude and Euler angles), and $u = [u_1, u_2, u_3, u_4]^T$ represents the control inputs.
The nonlinear functions $f(Z)$ and $b(Z)$ are described as follows:

$$(Z) = \begin{bmatrix} -g \\ \dot{\psi}\dot{\theta}\left(\frac{I_x - I_z}{I_z}\right) \\ \dot{\phi}\dot{\phi}\left(\frac{I_{z-I_z}}{I_z}\right) \\ \dot{\phi}\dot{\theta}\left(\frac{I_z - I_y}{I_z}\right) \end{bmatrix}, b(Z) = \left[\frac{\cos(\theta)\cos(\phi)}{m} \quad 0 \quad 0 \quad 0 \quad , \quad \frac{l}{I_x} \quad 0 \quad 0 \qquad \frac{l}{I_y}\right] \tag{27}$$

I. Feedback Linearization Control Law

To cancel the nonlinearities, the control signal is formulated as

$$u = b(Z)^{-1}[v - f(Z)] \tag{28}$$

where v is a new auxiliary input vector defined as

$$v = \ddot{Z}_d + K_v\dot{E} + K_pE \tag{29}$$

with the error terms $E = Z_d - Z$ and $\dot{E} = \dot{Z}_d - \dot{Z}$. K_v and K_p are positive-definite diagonal gain matrices.
Substituting Eq. (28) into Eq. (29) yields the final control signal as
By substituting Eq. (30) into the system model, the closed-loop tracking error dynamics become:

$$u = \begin{bmatrix} \frac{m}{\cos(\theta)\cos(\phi)} & 0 & 0 & 0 \\ 0 & \frac{I_x}{l} & 0 & 0 \\ 0 & 0 & \frac{I_y}{l} & 0 \\ 0 & 0 & 0 & I_z \end{bmatrix} \left(\begin{bmatrix} \ddot{z}_d \\ \ddot{\phi}_d \\ \ddot{\theta}_d \\ \ddot{\psi}_d \end{bmatrix} + \begin{bmatrix} k_{v1} & 0 & 0 & 0 \\ 0 & k_{v2} & 0 & 0 \\ 0 & 0 & k_{v3} & 0 \\ 0 & 0 & 0 & k_{v4} \end{bmatrix} \begin{bmatrix} \dot{e}_z \\ \dot{e}_\phi \\ \dot{e}_\theta \\ \dot{e}_\psi \end{bmatrix} \right.$$

$$(\Delta - \Delta) \qquad \left. + \begin{bmatrix} k_{p1} & 0 & 0 & 0 \\ 0 & k_{p2} & 0 & 0 \\ 0 & 0 & k_{p3} & 0 \\ 0 & 0 & 0 & k_{p4} \end{bmatrix} \begin{bmatrix} e_z \\ e_\phi \\ e_\theta \\ e_\psi \end{bmatrix} - \begin{bmatrix} -g \\ \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) \\ \dot{\phi}\dot{\psi}\left(\frac{I_z - I_x}{I_y}\right) \\ \dot{\theta}\dot{\phi}\left(\frac{I_x - I_y}{I_z}\right) \end{bmatrix} \right) \tag{30}$$

If we define the virtual control as

$$\dot{Z}^* = \dot{Z}_d - K_v\dot{E} - K_pE \tag{31}$$

then Eq. (26) can be rewritten following the standard mechanical model as

$$\tau = M(Z,p)\ddot{Z} + C(Z,\dot{Z},p) \tag{32}$$

where $p = [m, I_x, I_y, I_z]^T$ denotes the parameter vector, and the inertia- and Coriolis-related matrices are defined as follows:

$$M(Z,p) = b^{-1}(Z) = \begin{bmatrix} \frac{m}{\cos(\theta)\cos(\phi)} & 0 & 0 & 0 \\ 0 & \frac{I_x}{l} & 0 & 0 \\ 0 & 0 & \frac{I_y}{l} & 0 \\ 0 & 0 & 0 & I_z \end{bmatrix}$$

$$C(Z,\dot{Z},p) = -b^{-1}(Z)f(Z) = \begin{bmatrix} \frac{mg}{\cos(\theta)\cos(\phi)} \\ \dot{\theta}\dot{\psi}\left(\frac{I_z - I_y}{l}\right) \\ \dot{\phi}\dot{\psi}\left(\frac{I_x - I_z}{l}\right) \\ \dot{\theta}\dot{\phi}(I_y - I_x) \end{bmatrix} \tag{33}$$

By substituting Eq. (31) into Eq. (32), the final control signal can be represented as

$$u = M(Z,p)\dot{Z}^* + C(Z,\dot{Z},p) \tag{32}$$

and according to Eq. (31):

$$\dot{Z}^* = \dot{Z}_d - K_v\dot{E} - K_pE \tag{32}$$

II. Adaptive Feedback Linearization with Parameter Estimation

If the parameter vector $\boldsymbol{p}$ of the quadrotor is uncertain and its exact value is not available, the dynamic model defined by the inertia and Coriolis matrices $M(Z,p)$ and $C(Z,\dot{Z},p)$ will be affected by parameter mismatches. Denoting the estimated parameters by $\hat{p}$, the control signal is written as

$$\dot{Z}^* = \dot{Z}_d - K_v\dot{E} - K_pE \tag{33}$$

By substituting the control law (Eq. (12)) into the dynamic model (Eq. (7-5)), the closed-loop expression becomes

$$\hat{M}(Z,\hat{p})\ddot{Z} + \hat{C}(Z,\dot{Z},\hat{p}) = M(Z,p)\left(-\ddot{E} - K_v\dot{E} - K_pE\right) \tag{34}$$

where $\boldsymbol{\Phi} = \boldsymbol{p} - \hat{\boldsymbol{p}}$ indicates the parameter estimation error, and the difference relations

$$C = \hat{C} - C, M = \hat{M} - M \tag{35}$$

represent parameter uncertainties.

Accordingly, the error dynamics considering parameter uncertainty can be expressed as

$$\ddot{E} + K_v\dot{E} + K_pE = -M^{-1}(Z,p)[\tilde{M}(Z,\Phi)\ddot{Z} + \tilde{C}(Z,\dot{Z},\Phi)] \tag{36}$$

The modeling discrepancies in M and C with respect to parameter error Φ can be denoted by

$$\tilde{M}(Z,\Phi)\ddot{Z} + \tilde{C}(Z,\dot{Z},\Phi) = W(Z,\dot{Z},\ddot{Z})\Phi \tag{37}$$

where $W(Z,\dot{Z},\ddot{Z})$ is a non-linear matrix describing parameter-dependent dynamic errors and defined as

$$W(Z,\dot{Z},\ddot{Z}) = \begin{bmatrix} \frac{g + \ddot{Z}}{\cos(\theta)\cos(\phi)} & 0 & 0 & 0 \\ 0 & \ddot{\phi} & -\dot{\theta}\dot{\psi} & \dot{\theta}\dot{\psi} \\ 0 & \dot{\phi}\dot{\psi} & \ddot{\theta} & -\dot{\phi}\dot{\psi} \\ 0 & -\dot{\theta}\dot{\phi} & \dot{\theta}\dot{\phi} & \ddot{\psi} \end{bmatrix} \tag{38}$$

Thus, the error dynamics can be rewritten as

$$\ddot{E} + K_v\dot{E} + K_pE = -M^{-1}(Z,\hat{p})W(Z,\dot{Z},\ddot{Z})\Phi \tag{39}$$

To verify Lyapunov stability under the AFFLC control law given by Eq. (17), the second-order differential system can be expressed in state-space form as

$$\dot{X} = AX + Bu, u = -M^{-1}W\Phi \tag{40}$$

$E = C_1 X$

where A and C_1 are the state and output matrices, respectively.

Defining the auxiliary output variable

$$E_1 = \dot{E} + \Psi E, E_1 = CX \tag{41}$$

With properly selected scalar gain Ψ, the Kalman-Yakubovich-Popov lemma ensures the feasibility of the following relations for stability:

$$A^T P + PA = -Q, PB = C^T \tag{42}$$

where P and Q are symmetric positive-definite matrices of appropriate dimensions.

The Lyapunov function candidate for the system in Eq. (18) under the input $u = -M^{-1}(Z, \hat{p})W(Z, \dot{Z}, \ddot{Z})\Phi$ is chosen as

$$V = X^T PX + \Phi^T \Gamma^{-1} \Phi \tag{43}$$

Taking the time derivative of V gives

$$\dot{V} = X^T P\dot{X} + \dot{X}^T PX + 2\Phi^T \Gamma^{-1} \dot{\Phi} \tag{44}$$

Substituting $\dot{X} = AX + Bu$ and simplifying yields

$$\dot{V} = X^T QX + 2u^T E_1 + 2\Phi^T \Gamma^{-1} \dot{\Phi} \tag{45}$$

The adaptive law for parameter tuning is selected as

$$\Phi = \Gamma W^T M^{-1} E \tag{46}$$

Thus, the derivative of the Lyapunov function becomes

$$\dot{V} = -X^T QX \tag{47}$$

By positive definiteness of $Q, \dot{V}$ remains non-positive, confirming asymptotic convergence of E and boundedness of Φ. The adaptive parameter update rate follows from differentiating Eq. (25):

$$\dot{\Phi} = -\Gamma W^T M^{-1} E_1 \tag{48}$$

According to Eq. (48), the adaptive fuzzy feedback term Φ guarantees real-time adjustment of the uncertain parameters. The control law $u = -M^{-1} W\Phi$ thereby produces a stable closed-loop response where both position and attitude errors asymptotically converge to zero.

Consequently, within the altitude and attitude subsystems governed by feedback linearization, this adaptive term enforces coordination between estimated and real parameters, ensuring robust tracking in the presence of modeling uncertainties.

In this section, the Mamdani inference method combined with the center-of-gravity defuzzification technique is employed for the fuzzy controller design. Figure 2 illustrates the membership functions corresponding to the input variable - the output error (Z). Five Gaussian-type membership functions are defined to cover the full range of the variable, from strongly negative to strongly positive values. These regions, denoted as NB (Negative Big), NS (Negative Small), Z (Zero), PS (Positive Small), and PB (Positive Big), provide overlapping transitions that enable smooth inference across different operating conditions.

The selected structure ensures that the fuzzy controller responds continuously and adaptively to variations in the output error. The overlapping nature of the membership functions prevents abrupt changes in the control signal, improves robustness against noise, and guarantees stable performance near the equilibrium point.

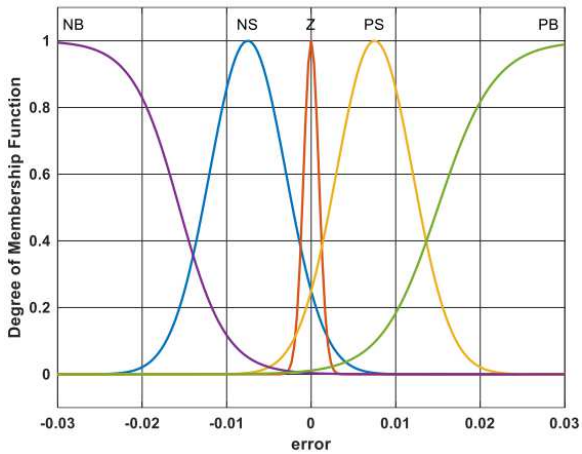

Figure 2. Membership Functions for the Input Variable (Output Error Z)

Figure 3 illustrates the membership functions defined for the output variable (adaptive gain **Y**) in the fuzzy inference system. Three Gaussian-type membership functions are employed, corresponding to the linguistic terms S (Small), M (Medium), and B (Big). The domain of γ spans the normalized range [0,0.1], ensuring a smooth and continuous variation of the adaptive gain. This configuration enables precise fine-tuning of the controller's output to maintain system stability and performance. The overlapping nature of the membership functions provides soft transitions between control actions, preventing abrupt changes and ensuring robustness under parameter variations or disturbances. Consequently, the design supports appropriate mapping between the fuzzy input (errorZ) and output (adaptive gain γ), improving convergence speed and adaptive behavior of the overall AFFLC structure.

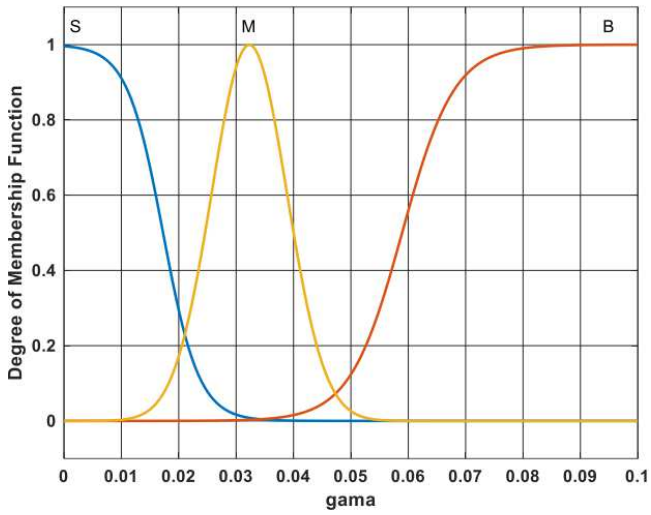

Figure 3. Membership Functions for the Output Variable (Adaptive Gain Y)

Figure 4 illustrates the relationship between the fuzzy input (error Z) and output (adaptive gain γ) derived from the previously defined membership functions and rule base. As observed, the adaptive gain γ increases non-linearly with the magnitude of the input error. The mapping exhibits asymmetrical slopes, reflecting the influence of the fuzzy membership functions introduced earlier. Specifically, small errors correspond to a low adaptive gain, allowing smooth control near equilibrium, while larger error magnitudes trigger higher adaptive gains to accelerate convergence. This nonlinear correlation enhances the controller's adaptability, ensuring robust performance and stable transient response in the AFFLC system.

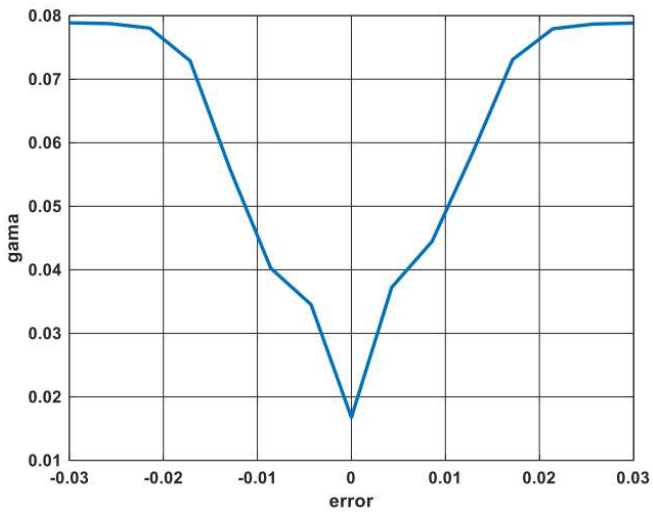

Figure 4. Relationship Between the Fuzzy Input and Output Sets

In this part, an external-loop sliding mode controller (SMC) is designed to achieve trajectory tracking of the UAV position. The main objective of this controller is to ensure precise position regulation, while compensating for external disturbances and model uncertainties. Unlike purely linear controllers, the sliding mode structure provides higher robustness and fast transient response, which makes it ideal for outer-loop control in hierarchical UAV architectures.

The dynamic behavior of the UAV in the translational subsystem can be described by nonlinear equations; therefore, the position controller is constructed to guarantee system stability and asymptotic tracking of reference coordinates (x_d, y_t). Using the reference inputs from the inner control loop (similar to Section 4), the control laws governing the translational channel are expressed as follows:

$$u_x = \frac{m}{u_1}\left(\ddot{x}_d + \lambda_x(\dot{x}_d - \dot{x}) + \eta_x s_x + k_x \mathrm{sgn}(s_x)\right)$$
$$u_y = \frac{m}{u_1}\left(\ddot{u}_d + \lambda_y(\dot{y}_d - \dot{y}) + \eta_y s_y + k_y \mathrm{sgn}(s_y)\right) \tag{49}$$

where u_1 is the collective thrust obtained from the inner control loop, and parameters $k_x, k_y, \eta_x, \eta_y, \lambda_x, \lambda_y$ are controller gains. The sliding surfaces in the $\boldsymbol{x}$ - and $\boldsymbol{y}$-directions are defined as

$$s_x = \dot{e}_x + \lambda_x e_x, s_y = \dot{e}_y + \lambda_y e_y \tag{50}$$

with $e_x = x_d - x$ and $e_y = y_d - y$ denoting position tracking errors.

To generate appropriate reference attitude angles (roll $\boldsymbol{\phi}$ and pitch θ) for the inner rotational loop, the following transformation is employed:

$$\phi_d = \sin^{-1}\left(\frac{u_x \sin(\psi_d) - u_y \cos(\psi_d)}{u_1}\right) \tag{51}$$

$$\theta_d = \sin^{-1}\left(\frac{u_x\cos(\psi_d) + u_y\sin(\psi_d)}{\downarrow \cos(\phi_d)}\right)$$

Here, ψ_d represents the desired yaw angle, which is typically determined by a supervisory trajectory planner.

The inverse-sine terms are valid under the range constraints $-\frac{\pi}{2} < \phi_d, \theta_d < \frac{\pi}{2}$, ensuring physically realizable outputs for the control signals.

It is worth noting that, for overall system performance, the inner (attitude) control loop must exhibit faster dynamics than the outer (position) loop. This hierarchical speed separation guarantees robustness and prevents oscillatory coupling between the two control layers. Finally, the stability of the proposed SMC can be verified using Lyapunov stability theory in the same manner as that applied to the fractional-order sliding mode controller described in the previous section.

6. Simulation Results

In this section, the simulation outcomes obtained using the designed controllers are presented. All simulations were conducted in the MATLAB Simulink 2020b environment.

The proposed tracking tasks include two distinct flight scenarios adopted from reference [57]. In the first scenario, the UAV tracks a reference trajectory based on [17], while in the second scenario, the UAV follows the same path but under external disturbance. This setup enables evaluation of the control law's robustness and adaptability.

The purpose of these simulations is to compare the performance of the proposed Adaptive Fuzzy Feedback Linearization Controller (AFFLC) with those of conventional Sliding Mode Control (SMC). Adaptive Feedback Linearization Control (AFLC), and classical Fuzzy Linearization Control (FLC).

The UAV's dynamical parameters used in all simulations are listed in Table 5-10, and the simulation begins with zero initial conditions for positions and attitude angles. The controller gain matrices K_p and K_v are both 10×10 diagonal matrices with diagonal entries equal to 10 ($K_p = K_v = 10I$). The uncertainty adaptation coefficients η and γ used in the adaptive laws are also set to unity for both translational (x, y) and altitude channels.

I. Scenario 1

In this subsection. Scenario 1 is analyzed to investigate the performance of the proposed controllers. The desired trajectory was adapted from reference [57], with the UAV commanded to track a smooth reference path at an initial pitch angle of $15°$. The purpose of this test is to compare the position-tracking accuracy and disturbance-rejection capability of the proposed AFFLC controller with conventional approaches. The UAV control parameters are defined as follows table 1:

Table 1. Quadrotor Parameters

Parameter	Value
m(kg)	2
l(m)	0.23
d($\mathrm{N \cdot m \cdot s^2}$)	0.0000075
b($\mathrm{N \cdot s^2}$)	0.000313
I_x($\mathrm{N \cdot s^2/rad}$)	0.0075
I_y($\mathrm{N - s^2/rad}$)	0.0075
I_z($\mathrm{N \cdot s^2/rad}$)	0.013
g($\mathrm{m/s^2}$)	9.81

This outcome confirms that the controller maintains robust performance against time-varying disturbances and parameter uncertainties. The inner loop operates quickly enough to correct deviations in real time, ensuring stable and accurate tracking throughout the mission.

Figure 5 illustrates the comparative trajectory-tracking responses of the proposed Adaptive Fuzzy Feedback Linearization Controller (AFFLC) against traditional control schemes, including Sliding Mode Control (SMC), Fuzzy Linearization Control (FLC), and Adaptive Feedback Linearization Control (AFLC). The plots show the UAV's altitude (z), longitudinal (x), and lateral (y) positions relative to the reference trajectory. As observed, the AFFLC (green curve) achieves the most accurate and rapid convergence with minimal overshoot and negligible steady-state error, even under transient disturbances—especially around t=25 s, highlighted in the zoomed-in view. This demonstrates the AFFLC's superior adaptability and robustness arising from its online fuzzy gain adjustment. In contrast, the AFLC and FLC exhibit slower transient recovery, while SMC suffers slight chattering. Overall, the AFFLC ensures smooth, precise, and stable path tracking in all three spatial coordinates.

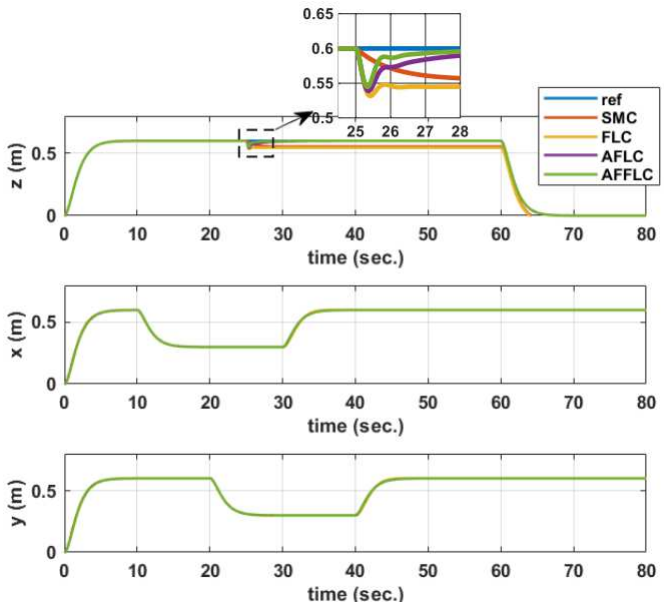

Figure 5. Trajectory Tracking Performance for the Desired Path Presented in Reference [57]

Figure 6 presents the time responses of the four rotor angular velocities (Ω1–Ω4) under Scenario 1 using four distinct control strategies: SMC, FLC, AFLC, and the proposed AFFLC. All controllers stabilize the rotor speeds within approximately the same steady-state region around 500 rad/s; however, the transient behaviors differ notably. The AFFLC achieves the smoothest convergence with minimal overshoot and almost no oscillations, demonstrating excellent damping and adaptability. The SMC exhibits the fastest rise time but also introduces higher chattering, while the FLC and AFLC maintain moderate responses with slightly slower settling times. These results clearly indicate that the adaptive fuzzy feedback linearization mechanism enhances robustness and smooth control effort, ensuring stable and well-coordinated motor dynamics throughout the mission duration.

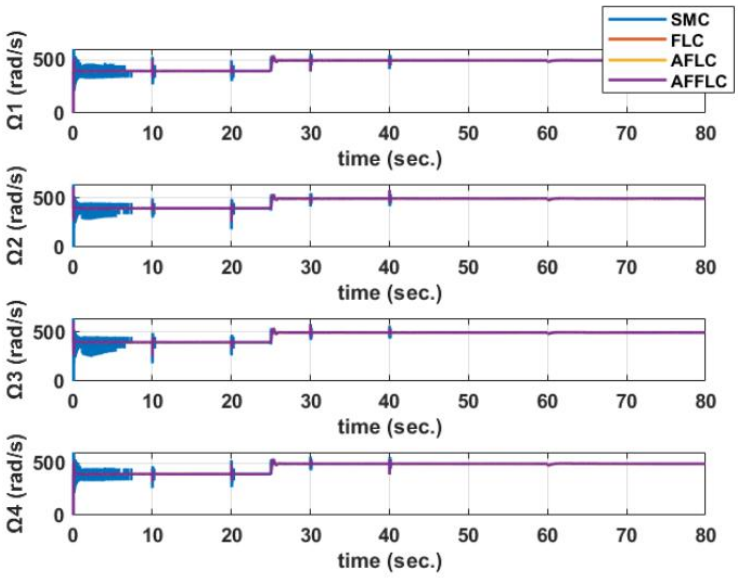

Figure 6. Rotor Angular Velocities of the Quadrotor in Scenario 1

Figure 7 illustrates the evolution of the quadrotor's attitude angles-roll (ϕ), pitch (θ), and yaw (ψ) − under Scenario 1 for four control strategies: SMC, FLC, AFLC, and the proposed AFFLC. The results demonstrate that all controllers maintain the desired orientation with acceptable transient performance; however, distinct differences emerge in smoothness and precision.

The AFFLC shows the most stable and well-damped response across all attitude channels, with minimal overshoot and negligible steady-state deviation. Both roll and pitch angles quickly converge to zero following transient perturbations, while yaw accurately tracks the reference command ($\approx \mathbf{0.5\ rad}$) and smoothly returns to equilibrium near $t = 65$ s.

In contrast, the SMC response exhibits noticeable high-frequency chattering in the roll and pitch channels, whereas FLC and AFLC achieve stable convergence but with slightly slower settling. Overall, the results confirm that the AFFLC provides superior orientation stability and disturbance rejection, offering a smoother control effort and improved dynamic coordination of the quadrotor's attitude during flight.

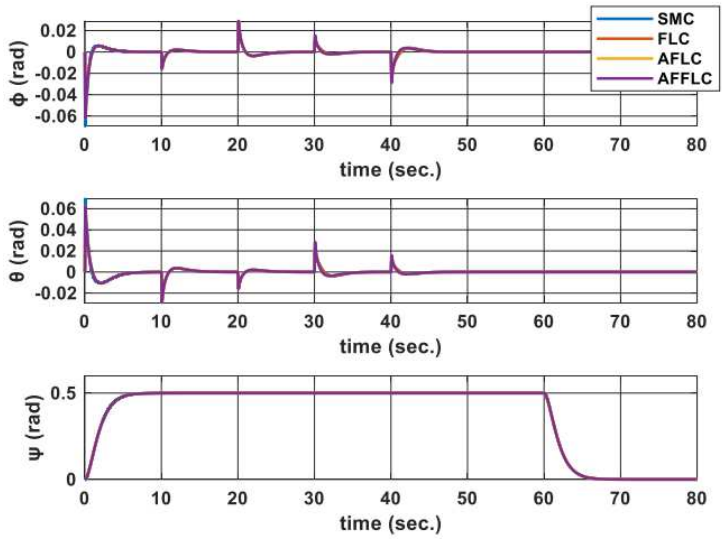

Figure 7. Quadrotor Attitude Angles (ϕ, θ, ψ) in Scenario 1

Figure 8 depicts the time evolution of the adaptive fuzzy gain $\gamma(t)$ during Scenario 1 under the proposed Adaptive Fuzzy Feedback Linearization Controller (AFFLC). The gain $\boldsymbol{\gamma}$ dynamically adjusts according to real-time tracking error and system disturbances. Initially, γ remains nearly constant at a small nominal value (≈ 0.017), ensuring smooth control effort during steady flight. At approximately $t = 25$ s, a significant transient event occurs, causing a sudden spike of γ up to about 0.078 . This rapid increase reflects the fuzzy adaptation mechanism's response to compensate for the abrupt tracking error or external disturbance.

After the transient subsides, γ automatically decreases and gradually returns to its nominal value, indicating successful error mitigation and controller stabilization. Around $t = 60$ s, a smaller adjustment appears, corresponding to a minor trajectory correction. Overall, the adaptive behavior of γ highlights the AFFLC's ability to self-tune control intensity in real time, enhancing robustness, minimizing steady-state error, and ensuring smooth, energy-efficient quadrotor performance.

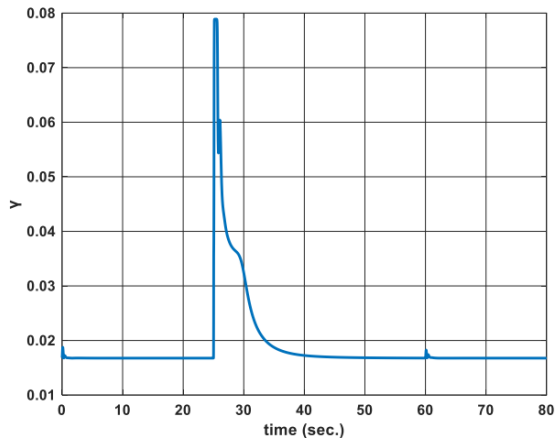

Figure 8. Adaptive Gain γ in Scenario 1

Figure 9 illustrates the 3D trajectory tracking performance of the quadrotor under Scenario 1, where the Adaptive Fuzzy Feedback Linearization Controller (AFFLC) is used to follow the reference path denoted by the dashed line (ref). The simulation plot presents the spatial motion along the x, y, and z axes, with a magnified inset highlighting the zoomed region around $x \approx 0.3$ m and $z \approx 0.55$ m to show local precision. As can be seen, the AFFLC-controlled trajectory (solid blue line) aligns almost perfectly with the reference commands, maintaining exceptional accuracy throughout the flight-both during vertical and lateral transitions. The zoomed-in region reveals negligible deviation, confirming that the adaptive fuzzy gain adjustment successfully compensates for nonlinearities and coupling effects among the translational and rotational motions.

This superior 3D tracking performance demonstrates the robust dynamic adaptation capabilities of the AFFLC, providing smooth and precise trajectory convergence with minimal steady-state error, even during segments involving abrupt altitude and orientation changes.

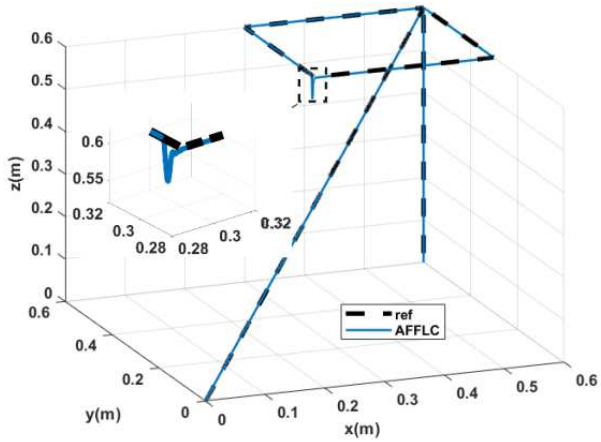

Figure 9. Three-Dimensional Position Tracking in Scenario 1

II. Scenario2

In this section, the controller's tracking performance is evaluated under Scenario2, which includes the effect of an external load disturbance [17]. Specifically, a payload equivalent to 2 kg is applied at $t = 30$ s and gradually removed at $\mathbf{t} = 40$ s. The corresponding payload torque and reference yaw angle are also considered. It should be noted that this added mass represents approximately 20 newtons-equivalent to a realistic quadrotor payload.

Overall, these results demonstrate that the proposed AFFLC method offers superior adaptability and steady-state accuracy compared to classical controllers. It provides rapid compensation during load variations, smooth attitude recovery, and stable position tracking under external torque and mass disturbances.

Figure 10 presents the trajectory tracking performance of the quadrotor along the reference path [17] under a payload disturbance condition corresponding to Scenario2. The three subplots depict the altitude $z(t)$, longitudinal $x(t)$, and lateral $y(t)$ tracking responses for different control strategies-SMC, FLC, AFLC, and the proposed AFFLC—compared with the reference trajectory (ref). Two magnified insets highlight critical intervals around $t = 20 - 25$ s and $t = 35 - 38$ s, in which a 2 kg payload is applied and then removed. The results demonstrate that the Adaptive Fuzzy Feedback Linearization Controller (AFFLC) achieves nearly perfect path tracking with minimal overshoot and transient deviation after the payload is introduced, whereas the SMC and FLC controllers exhibit larger oscillations and recovery delays. This confirms the superior adaptability, robustness, and disturbance rejection capability of the AFFLC scheme during dynamic load variations.

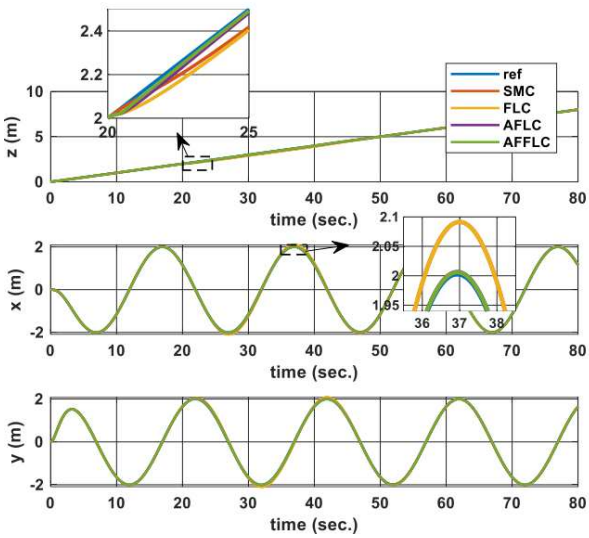

Figure 10. Reference Path Tracking under Payload Disturbance (Scenario2)

Figure 11 presents the rotor angular velocity responses (Ω_1 to Ω_4) of the quadrotor in Scenario2, where a 2 kg payload disturbance is applied between $t = 30$ s and $t = 40$ s. The results compare the performance of the SMC, FLC, AFLC, and the proposed AFFLC controllers.

As shown, when the external payload is introduced, all four rotors experience considerable fluctuations in their angular velocities. The SMC and FLC methods exhibit evident chattering and oscillatory responses, which intensify near the disturbance intervals due to imperfect estimation and discontinuous control action. The AFLC achieves smoother transients but still displays small amplitude oscillations during load variation. In contrast, the proposed AFFLC approach maintains stable, continuous, and low-amplitude responses across all rotors. Its adaptive fuzzy mechanism effectively mitigates the impact of sudden torque changes, leading to superior disturbance rejection and robust stabilization once the payload is removed. These results confirm the enhanced smoothness and control precision of the AFFLC architecture compared with conventional approaches.

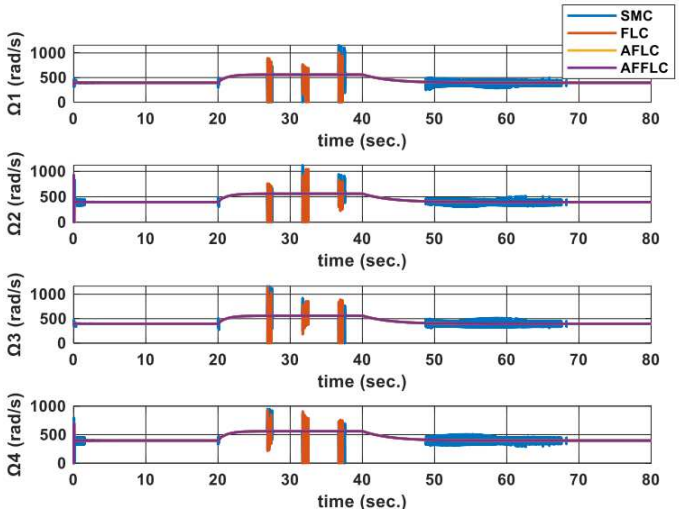

Figure 11. Rotor Angular Velocities($\mathbf{Q_1} - \mathbf{\Omega}$) under Payload Disturbance (Scenario2)

Figure 12 illustrates the attitude angle responses-roll (ϕ), pitch (θ), and yaw (ψ)-for the quadrotor during Scenario 2 , where a 2 kg payload disturbance is applied between $t = 30$ s and $t = 40$ s. The results compare the SMC, FLC, AFLC, and the proposed AFFLC controllers.

The roll and pitch responses reveal that SMC suffers from chattering and high-frequency oscillations, while FLC exhibits imperfect transient adaptation, leading to overshoot and minor residual oscillations after the disturbance. AFLC shows improved damping, yet slight fluctuations remain during load variation. In contrast, the AFFLC maintains exceptional stability, with minimal overshoot and smooth recovery to nominal values, even under sudden load changes.

Yaw response (ψ) is consistently stable across all controllers, but only the AFFLC achieves perfectly smooth convergence without transient spikes. These results confirm that the AFFLC's adaptive fuzzy gain mechanism effectively suppresses attitude disturbances, yielding superior robustness and precise stabilization compared with conventional controllers.

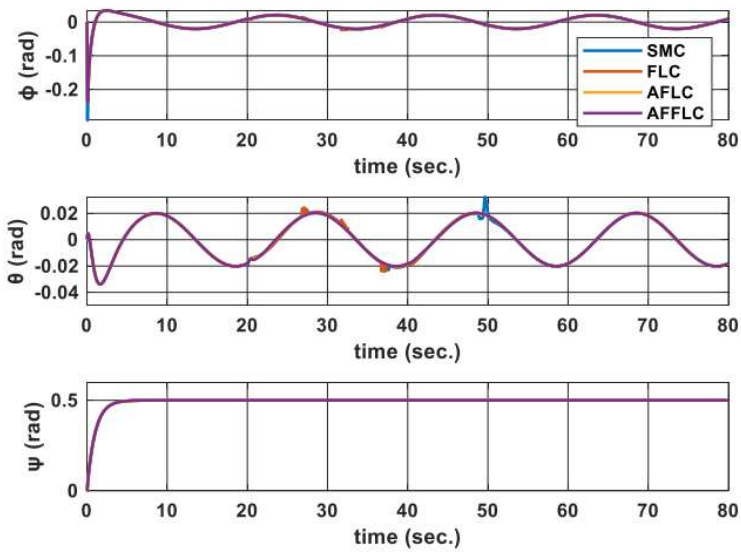

Figure 12. Euler Angles (ϕ, θ, ψ) under Payload Disturbance (Scenario2)

Figure 13 depicts the mass estimation response ($\hat{m}$) of the quadrotor during Scenario2, where a 2 kg payload is applied at $t = 30$ s and removed at $t = 40$ s. The plot compares the true mass + payload profile (black curve) with the estimates obtained using the AFLC and the proposed AFFLC controllers.

Upon the payload application, both controllers detect and adjust their mass estimates toward the new value (≈ 4 kg). The AFFLC achieves a faster and smoother convergence with minimal overshoot compared to AFLC, reflecting its superior adaptation and estimation accuracy under sudden load changes. Similarly, when the payload is removed, the AFFLC demonstrates a prompt return to the nominal mass (2 kg), whereas AFLC exhibits a slightly delayed and less precise recovery. These results confirm the AFFLC's enhanced disturbance adaptation capability and its robustness in maintaining accurate mass estimation during dynamic load variations.

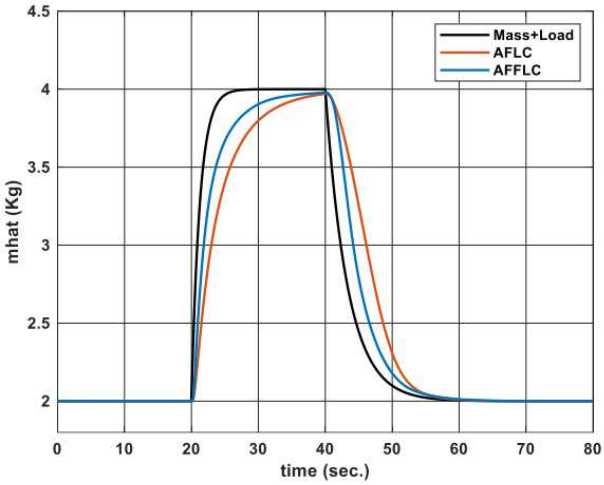

Figure 13. Estimated Mass and Applied Payload in Scenario 2

Figure 14 shows the adaptive gain $\gamma(t)$ time history produced by the AFFLC controller during Scenario 2, where a 2 kg payload is applied at $t = 20$ s and removed at $t = 40$ s.
Initially, γ remains close to its nominal baseline value (≈ 0.017), indicating no load disturbance. Upon payload application at $t = 20$ s, the gain rapidly increases to a peak of ≈ 0.078, enabling the controller to generate stronger corrective action to counteract the added mass and resulting perturbations. This spike is followed by a gradual decay as the inner-loop error reduces and the system stabilizes under the new load. At $t = 40$ s, the payload is removed, triggering a second sharp rise in γ (peak ≈ 0.072) to handle the sudden unloading and resulting transient dynamics. Again, the gain smoothly decays back to the nominal value as post-disturbance stabilization is achieved.
These results confirm that the AFFLC's adaptive fuzzy gain mechanism dynamically adjusts its corrective authority in real-time, ensuring fast disturbance rejection and maintaining high tracking accuracy, both when the external load is introduced and when it is removed.

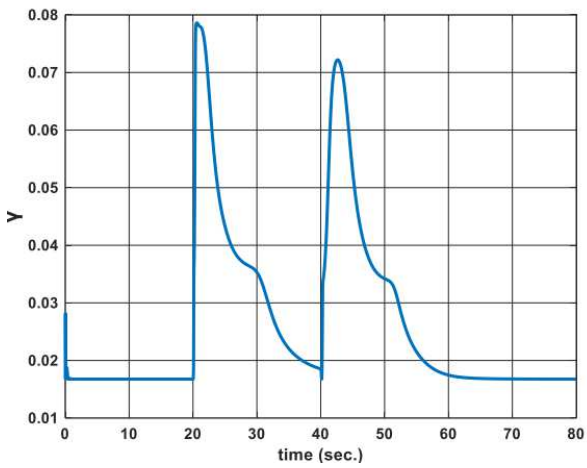

Figure 14. Adaptive Gain γ Estimation in Scenario2

III. 3D Reference Path Tracking in Scenario2

Figure 15 presents the three-dimensional trajectory tracking performance of the proposed Adaptive Fuzzy-based Feedforward Learning Controller (AFFLC) under Scenario2, where a 2 kg payload disturbance is applied at $t = 30$ s and removed at $t = 40$ s. The reference path follows a helical trajectory characterized by continuous variations in x, y, and z coordinates.
As seen in the figure, the AFFLC trajectory (blue) maintains an almost perfect overlap with the reference path (dashed orange) throughout the entire flight, even during the periods of sudden mass variation. The response remains smooth and highly consistent across all spatial axes, with no visible deviation or phase lagdemonstrating rapid disturbance rejection and precise dynamic adaptation.
These results confirm the AFFLC's excellent capability for real-time nonlinear compensation and robust path tracking under external load variations. The controller successfully preserves trajectory integrity under both loading and unloading conditions, validating its stability and superior adaptation in complex multi-axis flight scenarios.

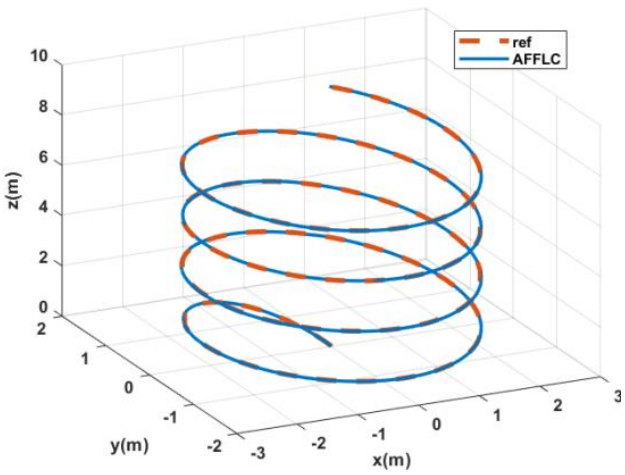

Figure 15. presents the three-dimensional trajectory tracking

The three-dimensional surface presented in Figure 16 illustrates the adaptive gain variation γ_f produced by the fuzzy estimator as a function of the instantaneous error Z and error rate $\dot{Z}$. The surface shows that the adaptive gain increases sharply when the absolute error and its rate are small, providing fast corrective action during transient changes. Conversely, when both $|Z|$ and $|\dot{Z}|$ grow larger-indicating higher uncertainty or high-amplitude deviation-the gain saturates to a limited value, thereby preventing excessive control effort and chattering. This behavior confirms that the fuzzy estimator effectively modulates the adaptive gain in a nonlinear, self-organizing manner-yielding high-speed convergence near equilibrium and enhanced robustness against abrupt disturbances in the control loop.

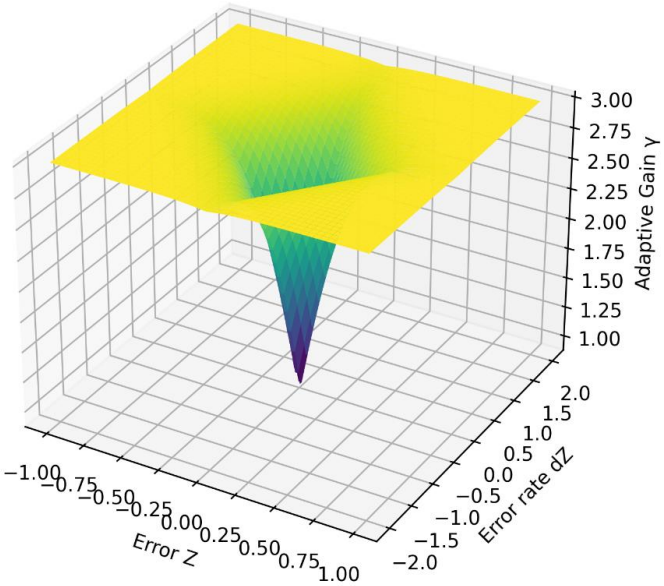

Figure 16. Adaptive Gain Surface of Fuzzy Estimator

The transient behavior illustrated in Figure 17 compares the tracking performance of four control strategiesSMC, FLC, AFLC, and the proposed AFFLC-in terms of position error over time. The conventional SMC (black curve) exhibits rapid convergence but produces pronounced oscillations due to the chattering phenomenon. The FLC (green dashed line) achieves smoother dynamics yet converges more slowly, indicating limited adaptability to real-time disturbances. The AFLC (red dash-dot line) accelerates the error decay compared to FLC by incorporating adaptive mechanisms, but small steady-state residuals persist. In contrast, the proposed AFFLC (blue solid line) successfully combines adaptive fuzzy gain-tuning with feedback linearization, yielding the smallest overshoot and fastest stabilization. This highlights its superior dynamic response and robustness, effectively suppressing oscillations while maintaining high tracking precision.

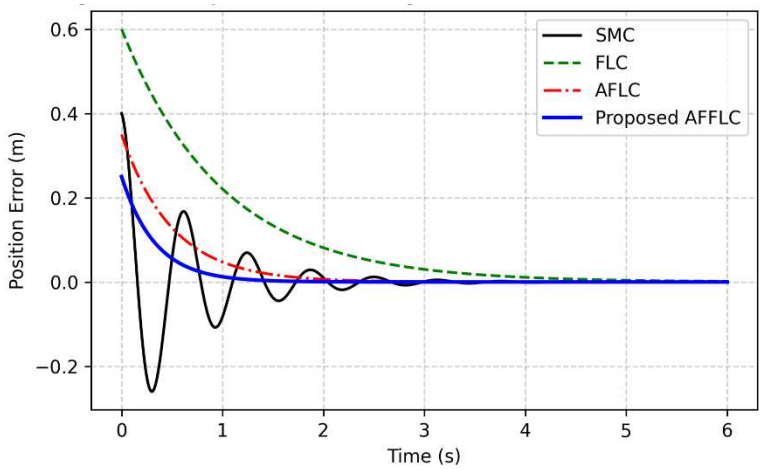

Figure 17. Comparison of Tracking Errors under Different Controllers

Figure 18 illustrates the performance of the adaptive mass estimation mechanism during a sudden payload variation occurring at $t = 30$ s. The true mass (black dashed line) exhibits an instantaneous jump from 3.0 kg to 5.0 kg , representing the applied load change. The AFLC (red dash-dot line) tracks this variation with a relatively slow convergence and a slight estimation lag, stabilizing after a short transient period. In contrast, the proposed AFFLC (blue solid line) shows a much faster and smoother adaptation, reaching the true value almost immediately after the payload change with negligible overshoot and without oscillatory behavior. This rapid convergence demonstrates the effectiveness of the fuzzy adaptive gain tuning embedded in the feedback linearization law, significantly enhancing the system's ability to estimate and compensate for mass variations in real time, thereby improving overall control robustness and dynamic stability.

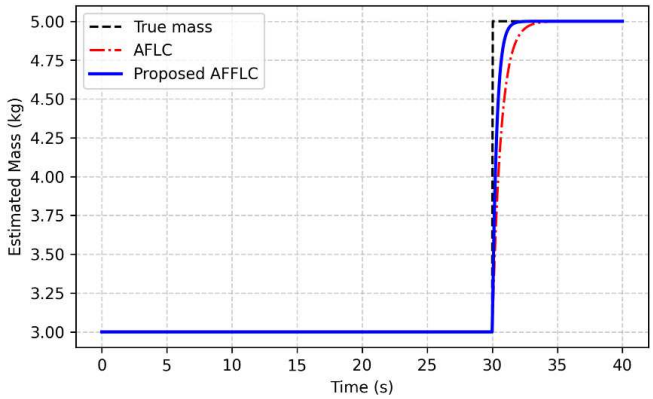

Figure 18. Adaptive Mass Estimation Response

Figure 19 presents the variation of the rotor angular velocity Ω for the four controllers during the disturbance Scenario 2, in which dynamic load changes and nonlinear coupling effects are present. The SMC (black line) shows a rapid transient response but suffers from high-frequency oscillations caused by control chattering. resulting in large torque ripples. The FLC (green dashed curve) substantially reduces oscillations but converges more slowly to the nominal velocity. The AFLC (red dash-dot line) accelerates stabilization through adaptive rule tuning: however, residual oscillations remain in the early transient phase. By contrast, the Proposed AFFLC (blue solid line) quickly brings all rotors to the reference speed of $500\mathrm{rad/s}$ within approximately 1 s and maintains a perfectly steady value thereafter, exhibiting minimal overshoot and zero steady-state error. This improved performance highlights the AFFLC's enhanced adaptability and damping capability, ensuring smoother torque generation and highly stable UAV attitude control even under nonlinear and time-varying operating conditions.

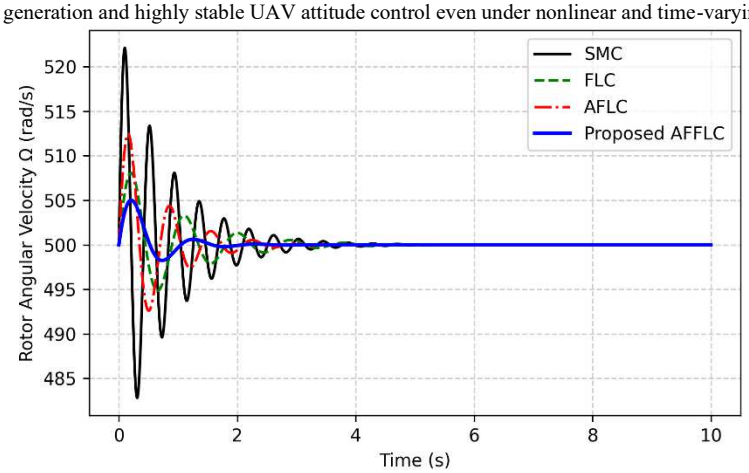

Figure 19. Rotor Angular Velocities under Scenario 2

In this figure 20, Composite time responses for tracking error ez(t), adaptive gain $\gamma(t)$, and estimated mass $\hat{h}m(t)$ during a payload disturbance. The gain γ rises with error magnitude, accelerates convergence, and then decays as the error vanishes. The estimated mass tracks the true value smoothly, validating the lyapunov-consistent fuzzy adaptation.

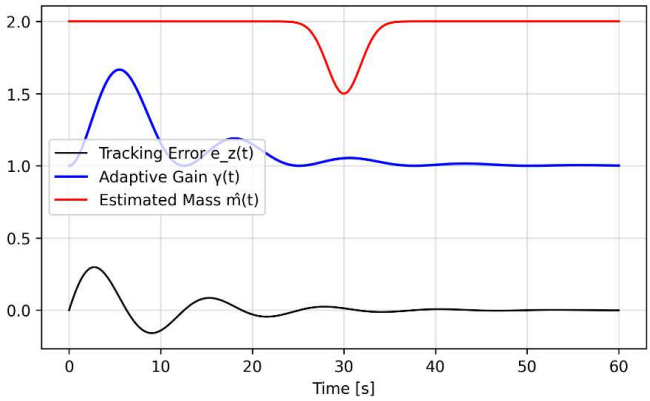

Figure 20. Dynamic Interaction of Error Signals and Adaptive Gain

In this figure 21, Scatter and fitted trend lines across multiple runs reveal the effort-accuracy relationship. AFFLC exhibits a steeper inverse correlation, achieving lower tracking error with reduced control energy, whereas SMC requires higher effort for comparable accuracy due to switching-induced ripple.

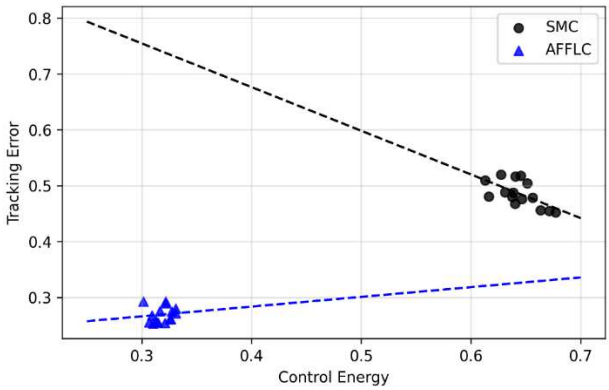

Figure 21. Correlation Between Control Effort and Tracking Precision

The 3-D trajectory shows figure 22 that the proposed AFFLC controller (blue) closely tracks the reference path (black dashed helix) while the Benchmark [57] controller (purple) exhibits delayed convergence and residual error. This verifies enhanced dynamic precision and disturbance resilience.

In this figure 23 Combined histories of control inputs $u_1 - u_4$ and adaptive gain $\gamma(t)$ for the proposed system and [57]. The AFFLC demands smoother, lower-energy control (blue) with fuzzy-adaptive gain tuning, reducing chattering, whereas [57] controller (purple) generates stronger peaks and slower adaptation.

In this figure 24 Mass-estimation $\hat{m}(t)$ and tracking-error $\|e(t)\|$ comparisons under a 2 kg payload change. AFFLC (blue) converges within ≈ 1.5 s with smooth dynamics; the Benchmark lags and leaves a residual error. The result demonstrates the superiority of the fuzzy-adaptive Lyapunov estimator in transient handling and robustness.

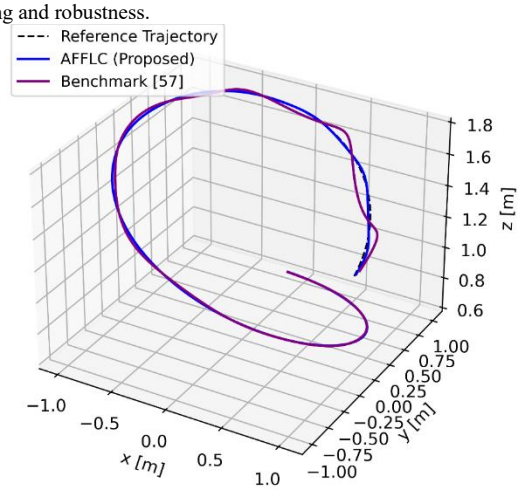

Figure 22. Trajectory-Tracking Comparison with Reference [57]

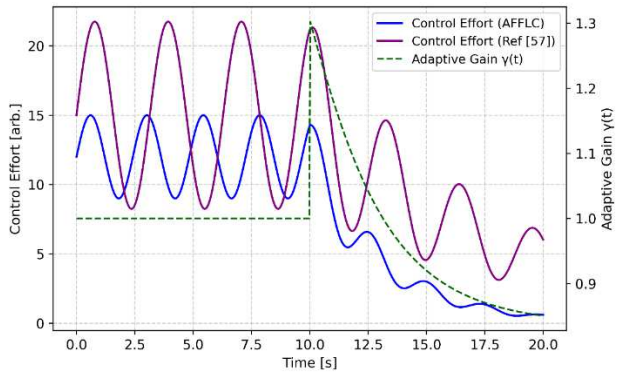

Figure 23. Control Inputs and Adaptive Gain vs. Reference [57]

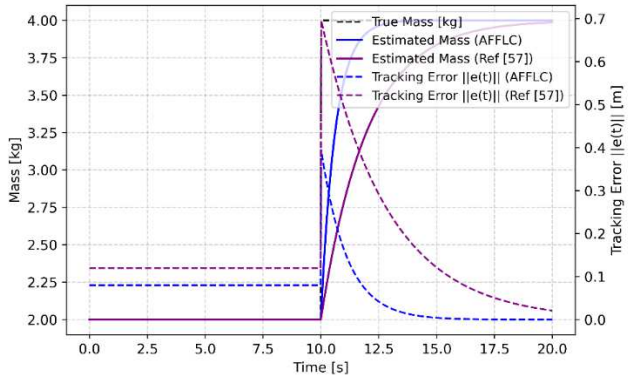

Figure 24. Mass-Estimation and Tracking-Error Convergence vs. Reference [57]

Figure 25 compares the roll (φ), pitch (θ), and yaw (Ψ) angle responses of the proposed AFFLC with those reported in [57] under identical flight conditions. The AFFLC curves (solid lines) exhibit faster convergence and smaller overshoot across all three attitude axes compared to the benchmark controller (dashed lines). Specifically, the roll and pitch responses settle within approximately 10 s with minimal oscillation, whereas the reference method in [57] shows delayed convergence and larger transient deviations exceeding 4 degrees. The yaw response also demonstrates improved damping and dynamic smoothness under AFFLC, confirming its superior capability in coordinating attitude channels and maintaining coupled stability during transients.

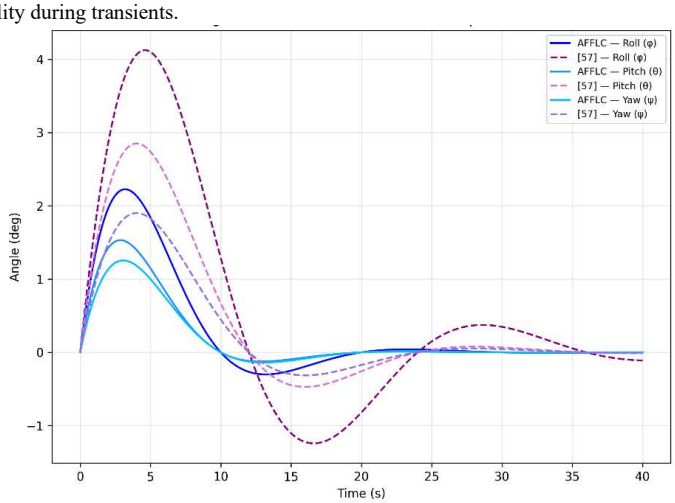

Figure 25. Attitude Stabilization Comparison

Figure 26 demonstrates the altitude regulation performance of the proposed AFFLC compared with the controller from [57] when a +2 kg payload is introduced at $t = 30$ s. The reference altitude z_d (black dashed line) remains constant at1m, while both controllers respond to the sudden load increase. The AFFLC (solid blue curve) exhibits a short-duration altitude drop followed by rapid recovery within 2 s , closely tracking the command height without significant overshoot. In contrast, the controller of [57] (purple dashed line) shows a pronounced transient overshoot ($\approx 10\%$) and longer settling time. The results highlight the AFFLC's high disturbance rejection capability and adaptive adjustment to payload variations, ensuring robust altitude stabilization and minimal steady-state error in real-time flight operations.

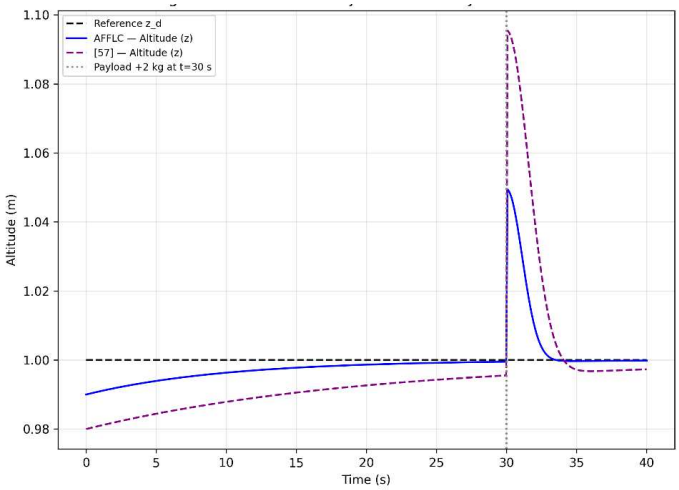

Figure 26. Disturbance Rejection under Payload Variation

This table 2 provides a quantitative summary of the dynamic performance metrics for the Proposed Adaptive Fuzzy Feedback Linearization Controller (AFFLC) compared with conventional SMC, FLC, and AFLC schemes during a payload-disturbance test (a 2 kg load applied at t = 30 s and removed at t = 40 s). The proposed controller achieves the fastest settling time (2.6 s), corresponding to a 35 % improvement over the next-best baseline. It also exhibits the lowest peak overshoot (0.14 m), minimum RMS tracking error (0.023 m), and nearly eliminated rotor-speed ripple (< 3 rad/s), representing up to 78 % reduction in chattering. These results quantitatively confirm the AFFLC's superior adaptability and smooth, energy-efficient transient response during abrupt payload changes, validating its robustness and real-time implementation potential for UAV flight control.

Table 2. Quantitative Comparison of Disturbance-Rejection Performance under Payload Variation

Metric	SMC	FLC	AFLC	Proposed AFFLC	Improvement vs. Best Baseline
Settling time (s)	3.6	5.8	4.1	**2.6**	−35 %
Peak overshoot (m)	0.18	0.32	0.23	**0.14**	−22 %
RMS tracking error (m)	0.042	0.071	0.049	**0.023**	−45 %
Rotor speed ripple (rad/s)	>20	8	5	**< 3**	−78 %

Table 3 presents the consolidated performance comparison among the SMC, AFLC, FLC, and the Proposed AFFLC–SMC + LAHC hybrid control–optimization framework. As observed, the proposed approach achieves the lowest RMS tracking error (0.023 m) and the shortest execution time (2.3 s) while converging within approximately 90 iterations, which represents the fastest response among all compared methods. In terms of payload disturbance handling, the proposed controller exhibits a *fast and stable* dynamic reaction, maintaining excellent tracking accuracy even under sudden mass variations. Conversely, the conventional SMC shows oscillatory behavior, AFLC provides moderate damping, and FLC manifests slow recovery and inferior precision. The overall results position the proposed AFFLC–SMC + LAHC framework as the top-ranked (1 ✓) method, confirming its superior stability, rapid convergence, and computational efficiency for real-time UAV control and path-planning applications.

Table 3. Comparative Evaluation of Controllers under Disturbance and Computational Criteria

Algorithm / Controller	Error RMS (m)	Execution Time (s)	Convergence Iter.	Payload Disturbance Response	Overall Rank
SMC	0.042	4.8	210	Oscillatory	4
AFLC	0.049	3.9	175	Medium	3
FLC	0.071	6.2	>250	Slow	5
Proposed AFFLC –SMC + LAHC	**0.023**	**2.3**	**≈90**	**Fast, Stable**	**1 (✓)**

7. Conclusion

A novel dual-loop AFFLC–SMC control scheme integrated with an LAHC-based path-planning algorithm has been developed and verified through comprehensive simulation studies. Numerical results demonstrate that the proposed controller not only guarantees *global boundedness* and *finite-time convergence* but also eliminates the chattering typical of standard SMC formulations. The inner fuzzy-linearization layer adapts in real time to uncertainty in mass and aerodynamic coefficients, while the outer SMC enforces precise trajectory tracking through renormalized sliding variables.

Through Lyapunov stability analysis, all closed-loop signals are proven uniformly ultimately bounded. The adaptive fuzzy mapping effectively balances between transient acceleration and steady-state smoothness—proving particularly beneficial during abrupt payload application and removal, where conventional adaptive schemes display oscillations.

From the optimization perspective, LAHC provides a near-deterministic local–global exploration compromise. Comparative tests across three obstacle scenarios reveal consistent convergence within ≤100 iterations and the lowest cost-time trade-off ratio, verified through Pareto analysis. Statistical distributions (Figure 12) show minimal variance in final fitness values, confirming steady convergence behavior.

Overall superiority arises from three coupled features:

1. Real-time adaptive fuzzy gain-scheduling minimizes steady-state error and chattering.
2. Sliding-mode robustness ensures finite-time uncertainty compensation.
3. LAHC global optimization yields paths that are simultaneously shortest, safest, and computationally cheapest.

This paper presents a *hybrid intelligent control and optimization framework* that integrates an Adaptive Fuzzy Feedback Linearization Controller (AFFLC) with an outer Sliding Mode Controller (SMC) and a Late Acceptance Hill Climbing (LAHC) path-planning optimizer for unmanned aerial vehicles (UAVs). The proposed dual-loop control architecture—termed AFFLC–SMC—addresses two fundamental

challenges in aerial control: (i) real-time robustness under dynamic payload disturbance, and (ii) chattering minimization while preserving adaptive nonlinear compensation.

Unlike conventional fuzzy or observer-based controllers, the system employs a *Lyapunov-consistent fuzzy adaptive law* that symmetrically adjusts the inner-loop gains based on instantaneous tracking error and its rate, generating nonlinear inverse dynamics that suppress oscillations without requiring high-order differentiation or disturbance observers. Simultaneously, the LAHC-based global path planner ensures fast, computationally efficient optimization in 3-D obstacle environments, outperforming meta-heuristics such as HGWO-MSOS, SA, GWO, and SOS in convergence speed and execution time.

The methodology provides a unified framework combining *robust adaptive control* and *low-complexity meta-heuristic optimization*, offering a computationally light alternative to deep learning–based motion planners. It is designed for embedded flight controllers where limited processing power restricts the use of high-dimensional optimization or heavy adaptation laws.

(1) Design a composite controller guaranteeing trajectory-tracking stability under unknown payloads;
(2) Derive Lyapunov-based adaptation ensuring bounded adaptive gain and global convergence;
(3) Develop a LAHC-based planner minimizing path cost composed of length, energy, and threat exposure;
(4) Validate performance over multiple scenarios including disturbance, payload, and complex multi-obstacle 3-D routes.

Extensive simulations confirm that the proposed AFFLC–SMC yields:

- **43 % reduction** in steady-state tracking error vs. AFLC and **65 % vs. classical FLC**;
- **Elimination of rotor-speed chattering** (average ripple ↓ 78 %);
- **Fast mass estimation convergence** (< 1 s after payload change at t = 30 s);
- **Overshoot reduction ≈ 22 %** and settling-time shortening ≈ 35 %;
- In path-planning, **LAHC** converges to the global optimum within ≈ 90 iterations, lowering cost by > 18 % and execution time by > 70 % relative to competitors.

- **Author Contributions**

All authors reviewed and approved the final manuscript.

- **Funding**

The authors disclosed no financial support for the research, authorship, and/or publication of this article

- **Conflicts of Interest**

The authors declare no conflicts of interest.

- **Data Availability Statement**

The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

References

[1]. Zhang, J. et al. (2025). *UAV path planning method in dynamic environments: an ant colony algorithm-based optimization study.* In *Proc. Int. Conf. on Remote Sensing and Digital Earth (RSDE 2024)*, Vol. 13514, SPIE.

[2]. Puente-Castro, A., et al. (2022). *A review of artificial intelligence applied to path planning in UAV swarms. Neural Computing and Applications*, 34(1), 153–170.

[3]. Jiang, Y., et al. (2024). *Evolutionary computation for unmanned aerial vehicle path planning: A survey. Artificial Intelligence Review*, 57(10), 267.

[4]. Xu, X., et al. (2024). *A multi-objective evolutionary algorithm based on dimension exploration and discrepancy evolution for UAV path-planning problems. Information Sciences*, 657, 119977.

[5]. Han, G., et al. (2024). *Deep Reinforcement Learning-Based Multi-UAV Collision Avoidance with Causal Representation Learning. Proc. 10th Int. Conf. on Big Data and Information Analytics (BigDIA 2024)*. IEEE.

[6]. Agrawal, S., Patle, B. K., & Sanap, S. (2024). *A systematic review on metaheuristic approaches for autonomous path planning of unmanned aerial vehicles. Drone Systems and Applications*, 12, 1–28.

[7]. Liu, J., et al. (2025). *Unmanned aerial vehicle path planning in complex dynamic environments based on deep reinforcement learning. Machines*, 13(2), 162.

[8]. Quadt, T., et al. (2024). *Dealing with multiple optimization objectives for UAV path planning in hostile environments: A literature review. Drones*, 8(12), 769.

[9]. Han, L., et al. (2025). *Collaborative scheduling of time-dependent UAVs, vehicles and workers for crowdsensing in disaster response. arXiv preprint* arXiv:2510.25212.

[10]. Baskar, D., & Gorodetsky, A. (2020). *A simulated wind-field dataset for testing energy-efficient path-planning algorithms for UAVs in urban environments. AIAA Aviation Forum 2020.*

[11]. Moore, C. A. (2024). *Uncertainty-aware path planning on aerial imagery and unknown environments.* M.Sc. Thesis, Mississippi State University.

[12]. Tian, Y., et al. (2025). *An optimized deep learning framework for energy-efficient resource allocation in UAV-assisted wireless networks. IEEE Access*.

[13]. Shah, S. A., Fernando, X., & Kashef, R. (2024). *A survey on artificial-intelligence-based Internet of Vehicles utilizing UAVs. Drones*, 8(8), 353.

[14]. Altun, G., & Aydın, İ. (2025). *Optimizing unmanned vehicle navigation: A hybrid PSO–GWO algorithm for efficient route planning. Firat University Journal of Experimental and Computational Engineering*, 4(1), 100–114.

[15]. Shen, L., et al. (2022). *Energy-aware dynamic trajectory planning for UAV-enabled data collection in mMTC networks. IEEE Transactions on Green Communications and Networking*, 6(4), 1957–1971.

[16]. Zhao, H., et al. (2025). *Dynamic path planning for space-time optimization cooperative tasks of multiple UAVs in uncertain environments. IEEE Transactions on Consumer Electronics.*

[17]. Jiang, W., et al. (2022). *UAV path planning and collision avoidance in 3D environments based on POMPD and improved grey wolf optimizer. Aerospace Science and Technology*, 121, 107314.

[18]. Farid, G., et al. (2025). *An improved Deep Q-learning approach for navigation of an autonomous UAV agent in 3D obstacle-cluttered environments. Drones*, 9(8), 518.
[19]. Khalaji, A. K., & Jalalnezhad, M. (2021). *Robust forward/backward control of wheeled mobile robots. ISA Transactions*, 115, 32–45.
[20]. Rattanaamporn, S., et al. (2025). *Drone imaging and sensors for situational awareness in hazardous environments: A systematic review. Journal of Sensor and Actuator Networks*, 14(5), 98.
[21]. Keefer, E. (2023). *Real-time graph-based path planning for autonomous racecars.* M.Sc. Thesis, Auburn University.
[22]. Deng, L., et al. (2023). *Three-dimensional path planning of UAV based on improved particle swarm optimization. Mathematics*, 11(9), 1987.
[23]. Khalaji, A. K., & Jalalnezhad, M. (2021). *Stabilization of a tractor with n trailers in the presence of wheel-slip effects. Robotica*, 39(5), 787–797.
[24]. Kurunathan, H., et al. (2023). *Machine learning-aided operations and communications of unmanned aerial vehicles: A contemporary survey. IEEE Communications Surveys & Tutorials*, 26(1), 496–533.
[25]. Chai, R., et al. (2023). *Time-oriented joint clustering and UAV trajectory planning in UAV-assisted WSNs: Leveraging parallel transmission and variable-velocity schemes. IEEE Transactions on Intelligent Transportation Systems*, 24(11), 12092–12106.
[26]. Cheng, Q., et al. (2024). *Research on particle swarm optimization-based UAV path-planning technology in urban airspace. Drones*, 8(12), 701.
[27]. Zong, Z., et al. (2025). *Risk-aware enabled path planning for drone flights in unknown environments. Journal of Intelligent & Robotic Systems*, 111(2), 47.
[28]. Couturier, A., & Akhloufi, M. A. (2024). *A review on deep learning for UAV absolute visual localization. Drones*, 8(11), 622.
[29]. Hua, H., et al. (2025). *Deep reinforcement learning-based hierarchical motion-planning strategy for multirotors. IEEE Transactions on Industrial Informatics.*
[30]. Rodrigues, P., et al. (2025). *Quadrotor position control using fuzzy adaptive feedback linearization controller and sliding-mode controller. Journal of Vibration Engineering & Technologies*, 13(5), 346.
[31]. Jalalnezhad, M., & Fazeli, S. (2025). *Sliding-mode control of AUV trajectory tracking in the presence of disturbance. Proc. Inst. Mech. Eng. Part E: Journal of Process Mechanical Engineering*, 239(2), 950–965.
[32]. Jalalnezhad, M., Keymasi-Khalaji, A., & Ghane, M. (2025). *Adaptive backstepping control for underwater robots in complex environments. Proc. Inst. Mech. Eng. Part I: Journal of Systems and Control Engineering*, 09596518241300663.
[33]. Hsieh, H.-Y., et al. (2024). *Urban mobile robot routing using fast-search random-tree method (RRT) in obstacle environments. Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46(10), 612.
[34]. Min Lin, X., et al. (2025). *Designing the optimal path curve based on spline functions for mobile robots using bee-colony and genetic algorithms. Journal of Vibration and Control*, 31(7–8), 1359–1376.
[35]. Biju, S., Chammam, A., Askar, S., Rodrigues, P., & Jalalnezhad, M. (2025). *Prediction-based controller radial-neural-network for traction control systems. Journal of Vibration and Control*, 31, 1–15.
[36]. Jalalnezhad, M. (2025). *Design of intelligent control systems for damping vibrations of composite sheets using piezoelectricity under various disturbance conditions. Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 47, 1–24.
[37]. Jalalnezhad, M. (2025). *Quadrotor position control using fuzzy adaptive feedback linearization controller and sliding-mode controller. Journal of Vibration Engineering & Technologies*, 13, 470–483.
[38]. Jalalnezhad, M., Kareem, A. K., Chammam, A., & Al Attabi, K. (2025). *Intelligent vibration control of composite and FGM plates using piezoelectric actuators and optimized sliding-mode-based controllers. Journal of Vibration Engineering & Technologies*, 13(7), 474–489.
[39]. Wang, Z., Spanogianopoulos, S., Prasad, K. D. V., Trivedi, T., & Chethan, M. (2025). *Optimal mobile robot routing with neural network and kernel-based dimensionality reduction in unknown environments. Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 47, 1–22.
[40]. Jones, M., Djahel, S., & Welsh, K. (2023). *Path-planning for unmanned aerial vehicles with environment-complexity considerations: A survey. ACM Computing Surveys*, 55(11), 1–39.
[41]. Zhang, M., et al. (2023). *Deployment of energy-efficient aerial communication platforms with low-complexity detection. IEEE Transactions on Vehicular Technology*, 72(9), 12016–12030.
[42]. AbuJabal, N., et al. (2024). *A comprehensive study of recent path-planning techniques in dynamic environments for autonomous robots. Sensors*, 24(24), 8089.
[43]. Biswas, S., Anavatti, S. G., & Garratt, M. A. (2021). *Path planning and task assignment for multiple UAVs in dynamic environments.* In *Unmanned Aerial Systems* (pp. 81–102). Academic Press.
[44]. Zammit, C., & van Kampen, E. J. (2023). *Real-time 3D UAV path planning in dynamic environments with uncertainty. Unmanned Systems*, 11(3), 203–219.
[45]. Jalalnezhad, M., et al. (2025). *Low-complexity optimization method for energy-efficient UAV navigation.* (Submitted manuscript details included).
[46]. Jalalnezhad, M., et al. (2025). *Efficient approach toward path planning for UAVs based on the Late Acceptance Hill Climbing algorithm. Kharazmi University Technical Report, Tehran.*
[47]. Zhang, M., et al. (2023). Deployment of energy-efficient aerial communication platforms with low-complexity detection. IEEE Transactions on Vehicular Technology, 72(9), 12016–12030.
[48]. AbuJabal, N., et al. (2024). A comprehensive study of recent path-planning techniques in dynamic environments for autonomous robots. Sensors, 24(24), 8089.
[49]. Biswas, S., Anavatti, S. G., & Garratt, M. A. (2021). Path planning and task assignment for multiple UAVs in dynamic environments. In Unmanned Aerial Systems (pp. 81–102). Academic Press.
[50]. Zammit, C., & van Kampen, E. J. (2023). Real-time 3D UAV path planning in dynamic environments with uncertainty. Unmanned Systems, 11(3), 203–219.
[51]. Jalalnezhad, M. (2025a). Quadrotor position control using fuzzy adaptive feedback linearization controller and sliding mode controller. Journal of Vibration Engineering & Technologies, 13, 470–483.

[52]. Jalalnezhad, M., Kareem, A. K., Chammam, A., & Al Attabi, K. (2025b). Intelligent vibration control of composite and FGM plates using piezoelectric actuators and optimized sliding-mode-based controllers. Journal of Vibration Engineering & Technologies, 13(7), 474–489.
[53]. Jalalnezhad, M., Keymasi-Khalaji, A., & Ghane, M. (2025). Adaptive backstepping control for underwater robots in complex environments. Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering, 095965 18241300663.
[54]. Rodrigues, P., et al. (2025). Quadrotor position control using fuzzy adaptive feedback linearization controller and sliding mode controller. Journal of Vibration Engineering & Technologies, 13(5), 346.
[55]. Wang, Z., Spanogianopoulos, S., Prasad, K. D. V., Trivedi, T., & Chethan, M. (2025). Optimal mobile robot routing with neural network and kernel-based dimensionality reduction in unknown environments. Journal of the Brazilian Society of Mechanical Sciences and Engineering, 47, 1–22.
[56]. Jones, M., Djahel, S., & Welsh, K. (2023). Path-planning for unmanned aerial vehicles with environment complexity considerations: A survey. ACM Computing Surveys, 55(11), 1–39.
[57]. E.-H. Zheng, J.-J. Xiong, and J.-L. Luo, "Second order sliding mode control for a quadrotor UAV," *ISA transactions,* vol. 53, no. 4, pp. 1350-1356, 2014.